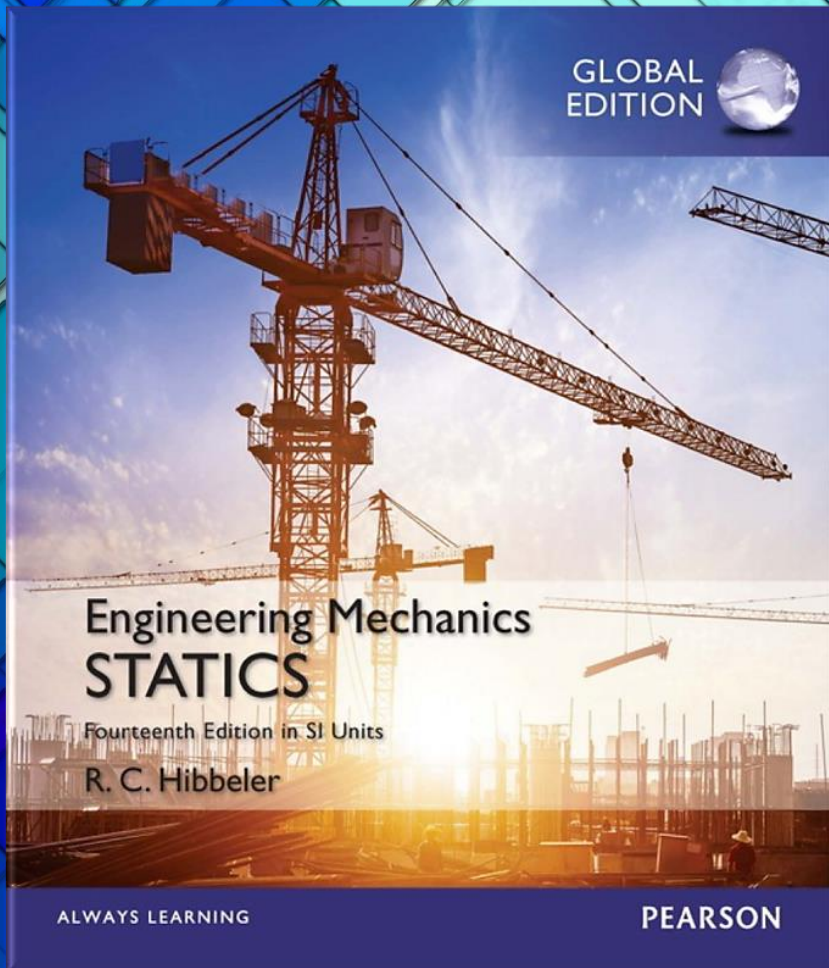
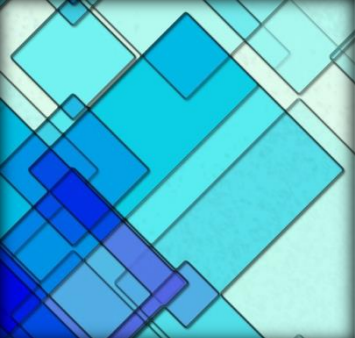


Engineering Mechanics: Statics

14th Edition
SI Units

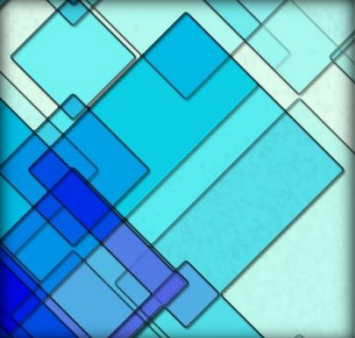


Chapter 8 Friction



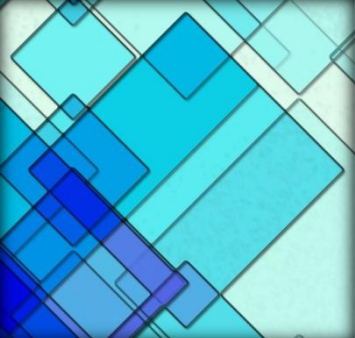
Chapter Objectives

- ❑ To introduce the concept of dry friction and show how to analyze the equilibrium of rigid bodies subjected to this force.
- ❑ To present specific applications of frictional force analysis on wedges, screws, belts, and bearings.
- ❑ To investigate the concept of rolling resistance.



Chapter Outline

- 8.1 Characteristics of Dry Friction**
- 8.2 Problems Involving Dry Friction**
- 8.3 Wedges**
- 8.4 Frictional Forces on Screws**
- 8.5 Frictional Forces on Flat Belts**
- 8.6 Frictional Forces on Collar Bearings, Pivot Bearings, and Disks**
- 8.7 Frictional Forces on Journal Bearings**
- 8.8 Rolling Resistance**



8.1 Characteristics of Dry Friction

Friction:

- ✓ Force that resists the movement of two contacting surfaces that slide relative to one another
- ✓ Acts tangent to the surfaces at the points of contact and is directed so as to oppose the possible or existing motion between the surface.
- ✓ Dry friction(Coulomb friction) – occurs when there is **no lubricating fluid**.



Figure: 08_PH001

The heat generated by the abrasive action of friction can be noticed when using this grinder to sharpen a metal blade.

Characteristics of Dry Friction

movement of two surfaces to slide relative to one another.

occurs at the points of contact as to oppose the direction of motion between the surfaces.

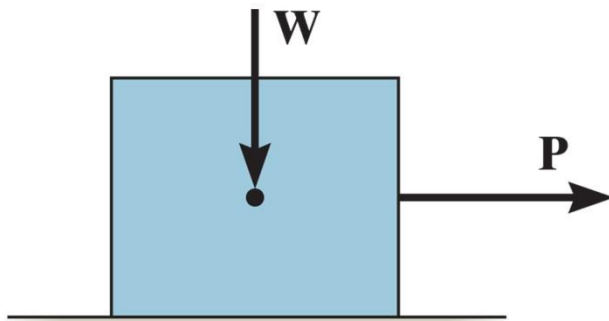
Static friction (no motion) – occurs when surfaces are not moving relative to each other.



8.1 Characteristics of Dry Friction

Theory of Dry Friction

- ✓ Consider the effects caused by pulling horizontally on a block of uniform weight W which is resting on a rough horizontal surface that is *nonrigid* or *deformable*.

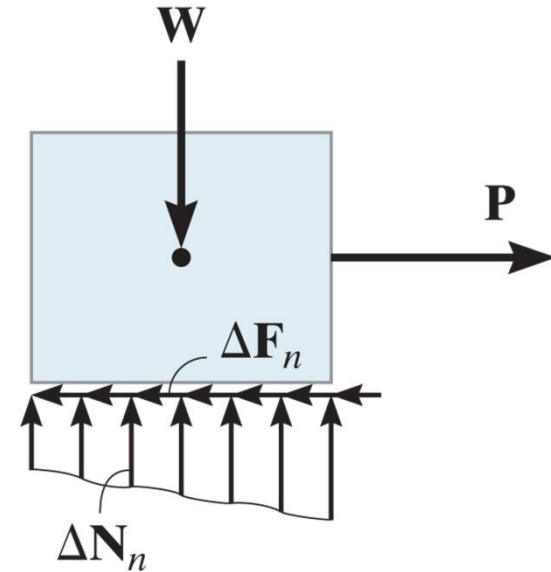


(a)

8.1 Characteristics of Dry Friction

Theory of Dry Friction

- ✓ Normal force ΔN_n and frictional force ΔF_n act along the contacting surface
- ✓ For equilibrium, normal forces act **upward** to balance the block's weight W ; frictional forces act to the left to prevent force P from moving the block to the right

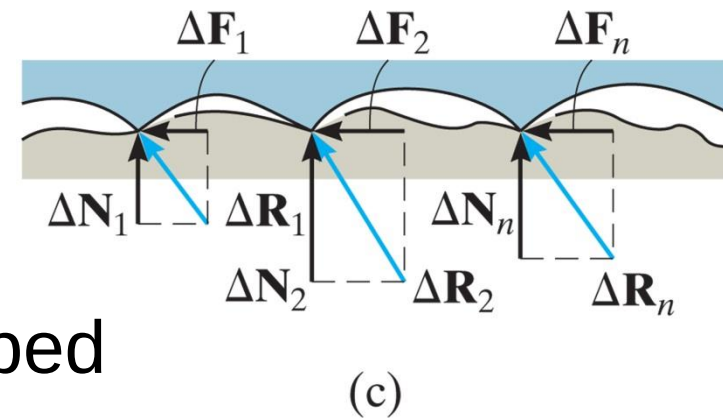


(b)

8.1 Characteristics of Dry Friction

Theory of Dry Friction

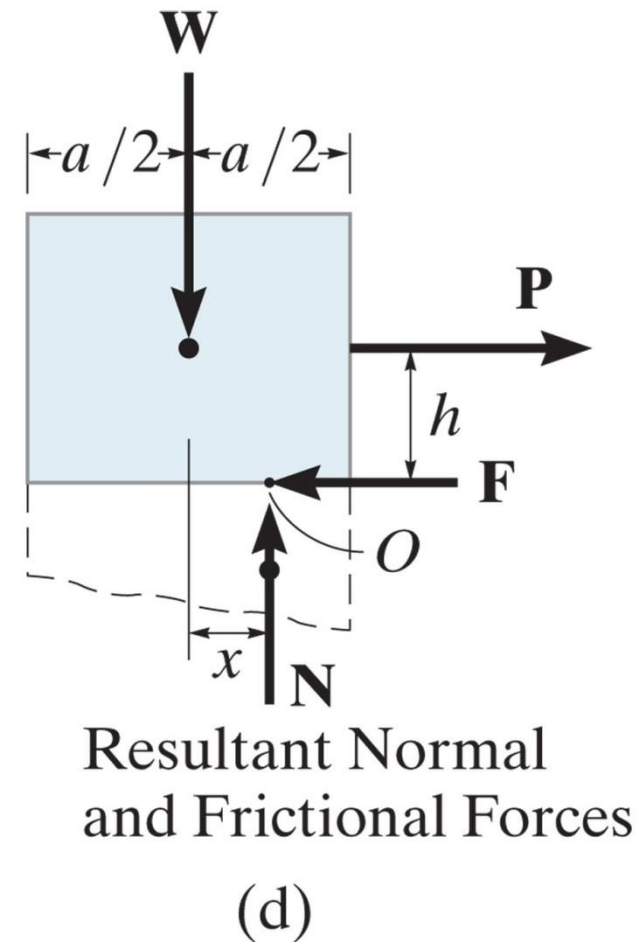
- ✓ Many microscopic irregularities exist between the two surfaces
- ✓ Reactive forces ΔR_n are developed at each point of contact.
- ✓ Each reactive force contributes both a frictional component ΔF_n and a normal component ΔN_n



8.1 Characteristics of Dry Friction

Equilibrium

- ✓ The effect of normal and frictional loadings is indicated by their resultant **N** and **F**
- ✓ **N** acts a distance x to the right of the line of action of **W**
- ✓ This location coincides with the centroid or the geometric center in order to balance the “tipping effect” caused by **P**



8.1 Chara

Equilibrium

- ✓ The effect of normal and loadings is indicated by resultant **N** and **F**
- ✓ **N** acts at a distance x to the line of action of **W**
- ✓ This location coincides with the centroid or the geometric center in order to balance the "effect" caused by **P**



Figure: 08_PH002

Regardless of the weight of the rake or shovel that is suspended, the device has been designed so that the small roller holds the handle in equilibrium due to frictional forces that develop at the points of contact,

A, B, C.

8.1 Characteristics of Dry Friction

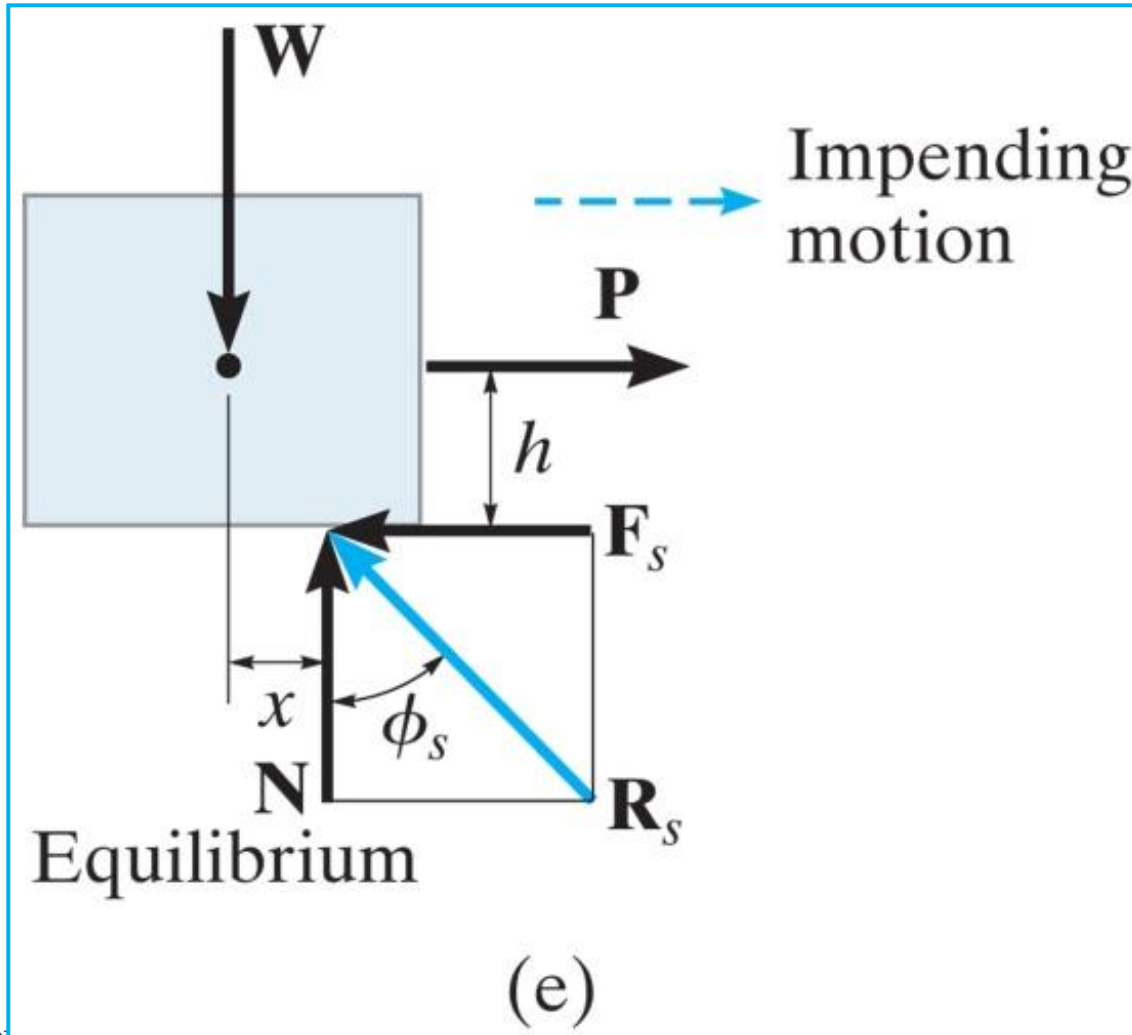
Impending Motion

- ✓ As P is slowly increased, F correspondingly increase until it attains a certain maximum value F_s , called the *limiting static frictional force*.
- ✓ When the block is in *unstable equilibrium*, further increase in P will cause the block to **move**.
- ✓ Limiting static frictional force F_s is directly **proportional** to the resultant normal force N

$$F_s = \mu_s N$$

μ_s : coefficient of static friction

8.1 Characteristics of Dry Friction



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8.1 Characteristics of Dry Friction

Impending Motion

- ✓ When the block is on the verge of sliding, N and F_s combine to create R_s
- ✓ Angle ϕ_s that $\mathbf{R}_s$ makes with $\mathbf{N}$ is called the angle of static friction

$$\phi_s = \tan^{-1} \left(\frac{F_s}{N} \right) = \tan^{-1} \left(\frac{\mu_s N}{N} \right) = \tan^{-1} \mu_s$$

8.1 Characteristics of Dry Friction

Impending Motion

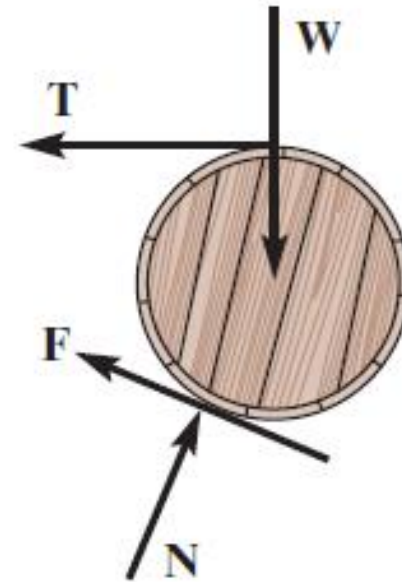
- ✓ When the block is on the verge of motion, F_s and F_s combine to form a resultant force R that acts at an angle ϕ_s to the normal force N .
- ✓ Angle ϕ_s that F_s makes with N is called the angle of static friction.

$$\phi_s = \tan^{-1} \mu_s$$

Table 8–1 Typical Values for μ_s

Contact Materials	Coefficient of Static Friction (μ_s)
Metal on ice	0.03–0.05
Wood on wood	0.30–0.70
Leather on wood	0.20–0.50
Leather on metal	0.30–0.60
Aluminum on aluminum	1.10–1.70

8.1 Characteristics of Dry Friction

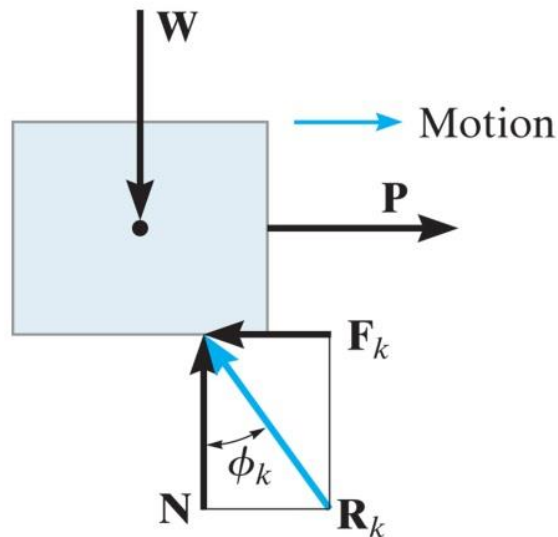


Some objects, such as this barrel, may not be on the verge of slipping, and therefore the friction force F must be determined strictly from the equations of equilibrium.

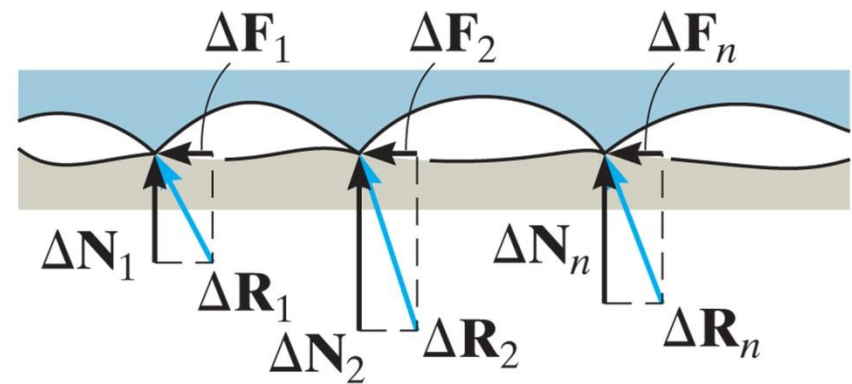
8.1 Characteristics of Dry Friction

Motion

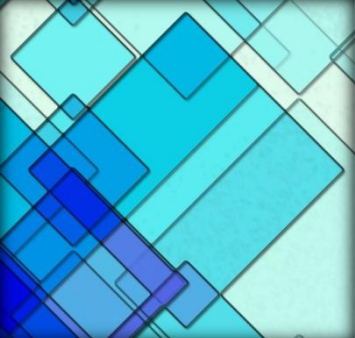
- ✓ When $\mathbf{P}$ is slightly greater than F_s , the frictional force will drop to a smaller value F_k , called **kinetic frictional force**.



(a)



(b)



8.1 Characteristics of Dry Friction

Motion

The magnitude of the kinetic friction force is directly **proportional** to the magnitude of the resultant force

$$F_k = \mu_k N$$

μ_k : coefficient of kinetic friction. 25% smaller than Table 8-1 for μ_s

8.1 Characteristics of Dry Friction

Motion

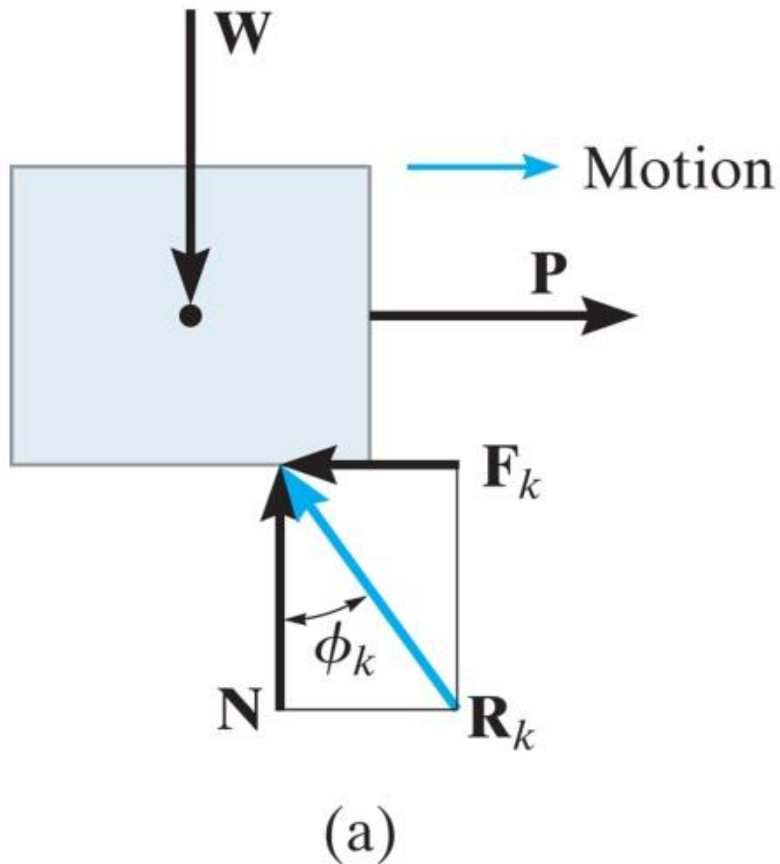
The magnitude directly **proporti** resultant force

μ_k : coeffici
smaller than

Table 8–1 Typical Values for μ_s

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8.1 Characteristics of Dry Friction



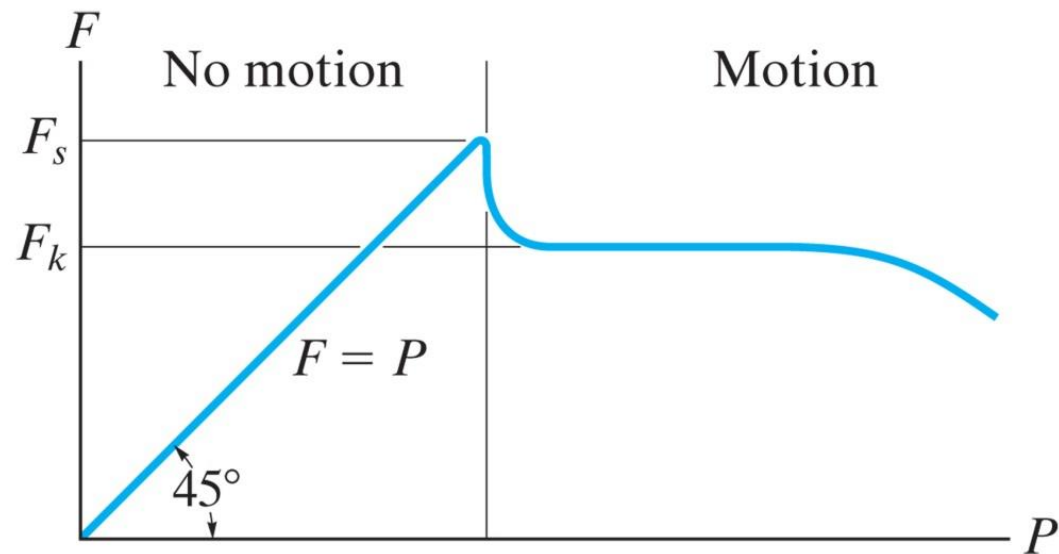
R_k is defined by ϕ_k . This angle is referred to as the angle of kinetic friction

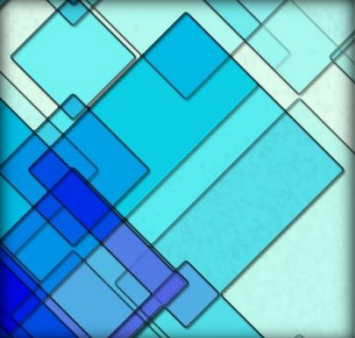
$$\phi_k = \tan^{-1}\left(\frac{F_k}{N}\right) = \tan^{-1}\left(\frac{\mu_k N}{N}\right) = \tan^{-1} \mu_k$$

$$\phi_s \geq \phi_k$$

8.1 Characteristics of Dry Friction

- ✓ F is a *static frictional force* if equilibrium is maintained.
- ✓ F is a *limiting static frictional force* F_s when it reaches a maximum value needed to maintain equilibrium.
- ✓ F is a *kinetic frictional force* F_k when sliding occurs at the contacting surface.

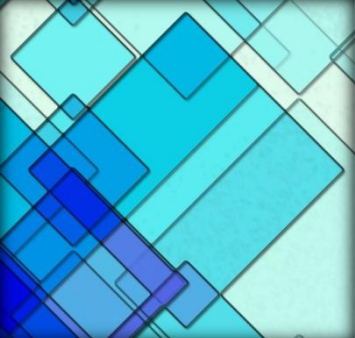




8.1 Characteristics of Dry Friction

Characteristics of Dry Friction

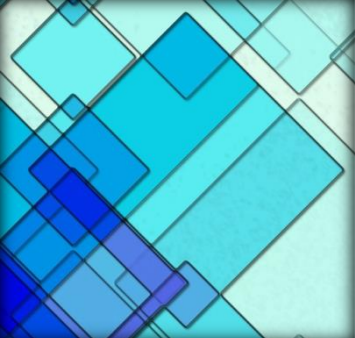
- The frictional force acts *tangent* to the contacting surfaces in a direction *opposed* to the *motion* or tendency for motion of one surface relative to another.
- The maximum static frictional force F_s that can be developed is independent of the area of contact, provided the normal pressure is not very low nor great enough to severely deform or crush the contacting surfaces of the bodies.



8.1 Characteristics of Dry Friction

Characteristics of Dry Friction

- The maximum static frictional force is generally greater than the kinetic frictional force for any two surfaces of contact. However, if one of the bodies is moving with a *very low velocity* over the surface of another, F_k becomes approximately equal to F_s , i.e., $\mu_s \approx \mu_k$.
- When *slipping* at the surface of contact is *about to occur*, the maximum static frictional force is proportional to the normal force, such that $F_s = \mu_s N$.
- When *slipping* at the surface of contact is *occurring*, the kinetic frictional force is proportional to the normal force, such that $F_k = \mu_k N$.

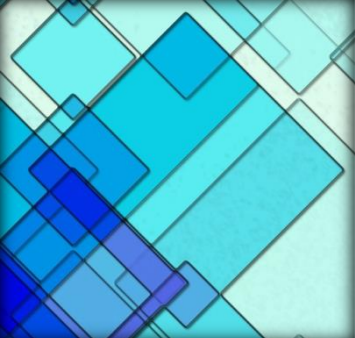


8.2 Problems Involving Dry Friction

Types of Friction Problems

3 types of mechanics problem involving dry friction:

1. No apparent impending motion
2. Impending motion at all points of contact
3. Impending motion at some points of contact

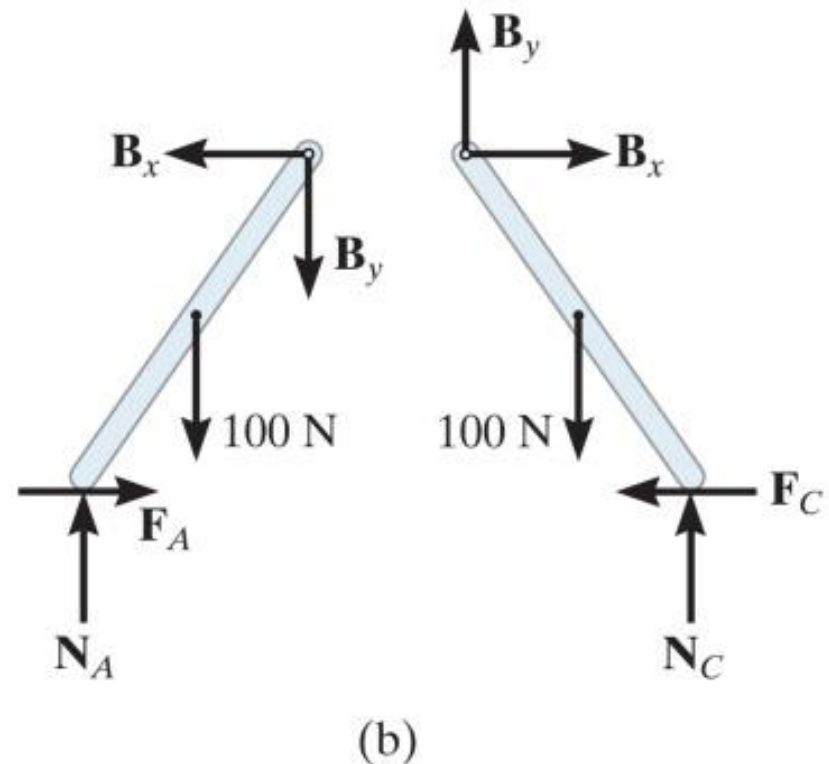
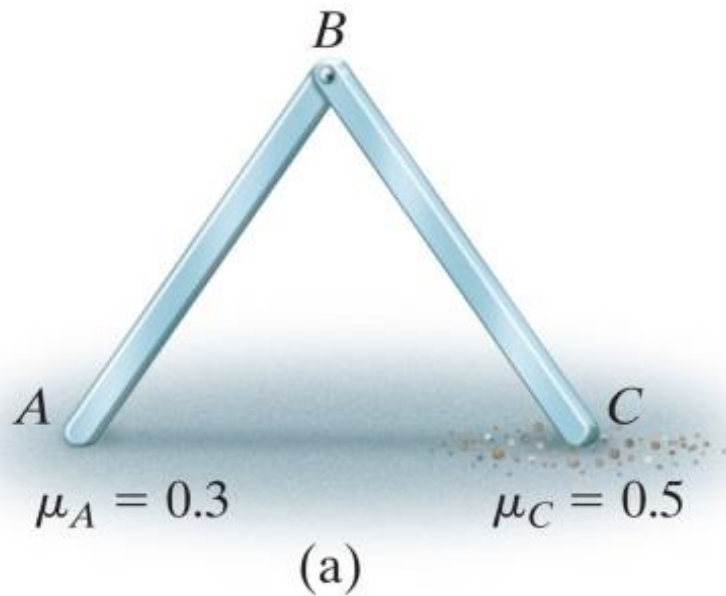


8.2 Problems Involving Dry Friction

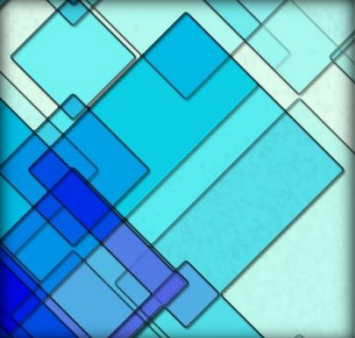
No Apparent Impending Motion (Equilibrium problems)

- ✓ Total number of unknowns = Total number of available equilibrium equations
- ✓ Frictional forces must satisfy $F \leq \mu_s N$; otherwise, slipping will occur and the body will not remain in equilibrium

8.2 Problems Involving Dry Friction



We must determine the frictional forces at A and C to check for equilibrium



8.2 Problems Involving Dry Friction

Impending Motion at All Points of Contact

- ✓ Total number of unknowns = Total number of available equilibrium equations plus the total number of available frictional equations, $F = \mu n$
- ✓ When motion is impending at the points of contact, then $F_s = \mu_s N$; whereas if the body is slipping, then $F_k = \mu_k N$.

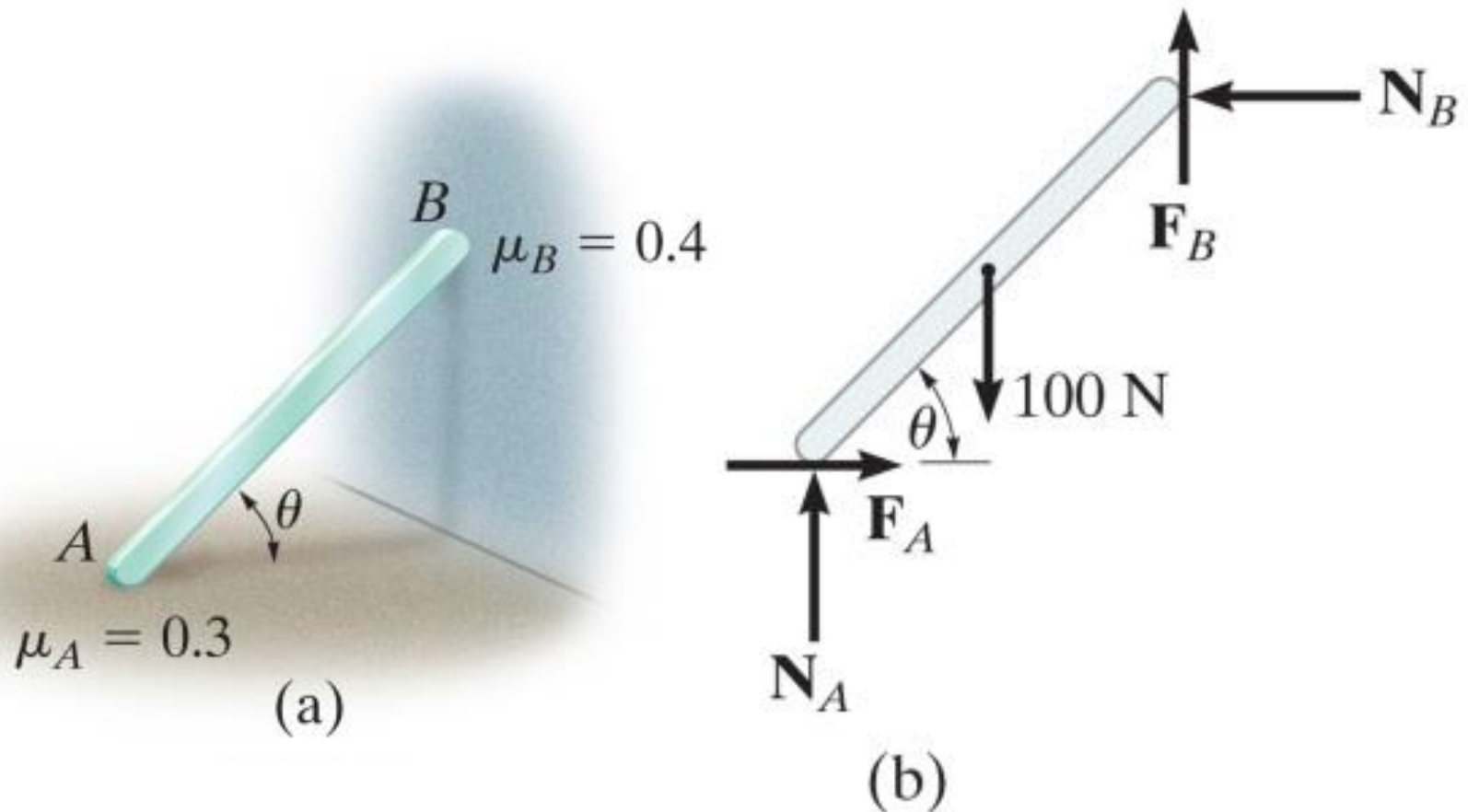
8.2 Problems Involving Dry Friction

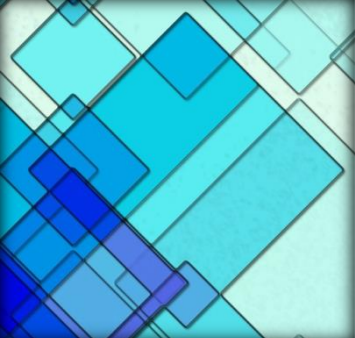
For example

- ✓ Consider the problem of finding the smallest angle Θ at which the 100-N bar in Fig. 8-5a can be placed against the wall without slipping.
- ✓ The **five** unknowns are determined from the **three** equilibrium equations and **two** static frictional equations which apply at both points of contact, so that $F_A = 0.3N_A$ and $F_B = 0.4N_B$.

8.2 Problems Involving Dry Friction

For example





8.2 Problems Involving Dry Friction

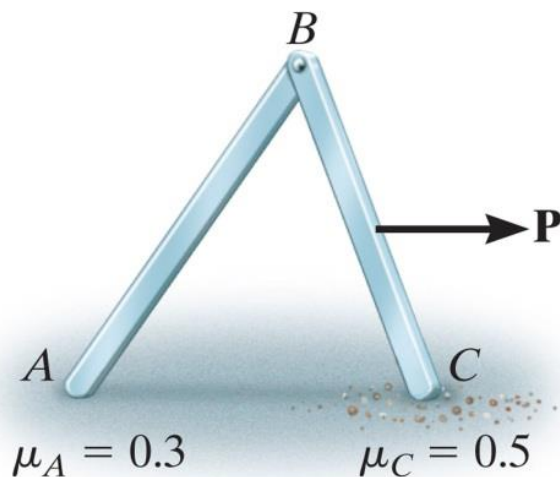
Impending Motion at Some Points of Contact

- ✓ The number of unknowns will be **less** than the number of available equilibrium equations plus the number of available frictional equations or conditional equations for tipping.
- ✓ As a result, several possibilities for motion or impending motion will exist and the problem will involve a determination of the kind of motion which actually occurs.

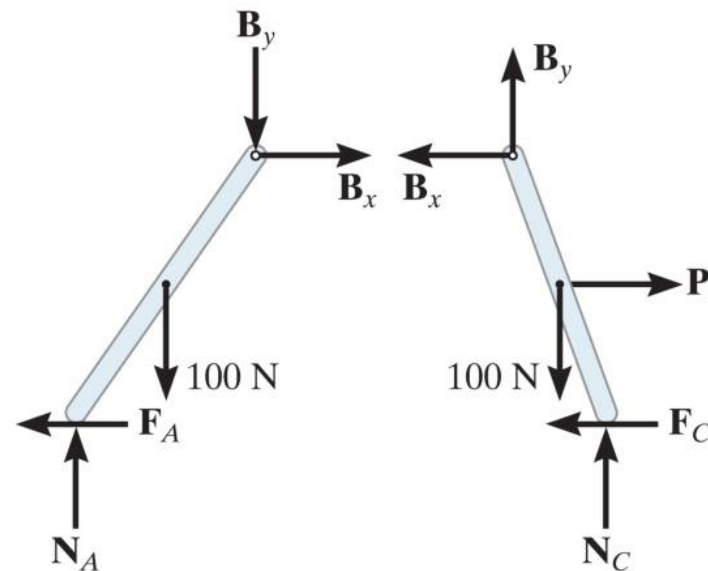
8.2 Problems Involving Dry Friction

In Fig. 8-6a, we wish to determine the horizontal force P needed to cause movement.

If each member has a weight of 100 N, then the FBD in Fig. 8-6b. There are **seven** unknowns.



(a)



(b)

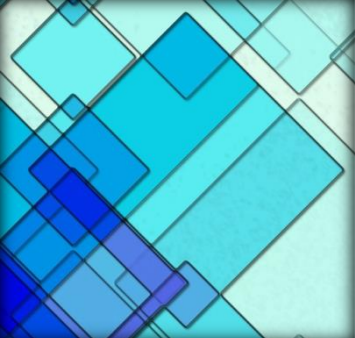
8.2 Problems Involving Dry Friction

- ✓ For a unique solution we must satisfy the six equilibrium equations (three for each member) and only **one** of two possible static frictional equations. This means that as P increases it will either cause slipping at A and no slipping at C , so that $F_A = 0.3N_A$ and $F_C \leq 0.5N_C$;
- ✓ Or slipping occurs at C and no slipping at A , in which case $F_C = 0.5N_C$ and $F_A \leq 0.3N_A$. The actual situation can be determined by calculating P for each case and then choosing the case for which P is smaller.
- ✓ If in both cases the same value for P is calculated, which would be highly improbable, then slipping at both points occurs simultaneously; i.e., **the seven unknowns** would satisfy **eight equations**.

8.2 Problems Involving Dry Friction

Equilibrium Versus Frictional Equations

- ✓ The friction force F is to be an “equilibrium force” and satisfies the inequality $F < \mu_s N \rightarrow$ assume the sense of direction of F on the FBD.
- ✓ The correct sense is made **after** solving the equilibrium equations
- ✓ If F is a negative scalar, the sense of $\mathbf{F}$ is the reverse of that assumed



8.2 Problems Involving Dry Friction

Equilibrium Versus Frictional Equations

- ✓ However, in cases where the frictional equation $F = \mu N$ is used in the solution of a problem, the convenience of assuming the sense of F is lost.
- ✓ F must always act with its correct sense on the FBD, whenever the frictional equation is used for the solution of a problem.

8.2 Problems Involving Dry Friction

Equilibrium Versus Friction

- ✓ However, in cases where $F = \mu N$ is used in the solution, the convenience of assuming $F = \mu N$ is often used.
- ✓ F must always act with its proper direction in the FBD, whenever the friction force is involved in the solution of a problem.



Depending upon where the man pushes on the crate, it will either tip or slip.

Procedure for analysis

Free-Body Diagrams.

- Draw the necessary free-body diagrams, and unless it is stated in the problem that impending motion or slipping occurs, *always* show the frictional forces as unknowns (i.e., *do not assume* $F = \mu N$).
- Determine the number of unknowns and compare this with the number of available equilibrium equations.
- If there are more unknowns than equations of equilibrium, it will be necessary to apply the frictional equation at some, if not all, points of contact to obtain the extra equations needed for a complete solution.
- If the equation $F = \mu N$ is to be used, it will be necessary to show $\mathbf{F}$ acting in the correct sense of direction on the free-body diagram.

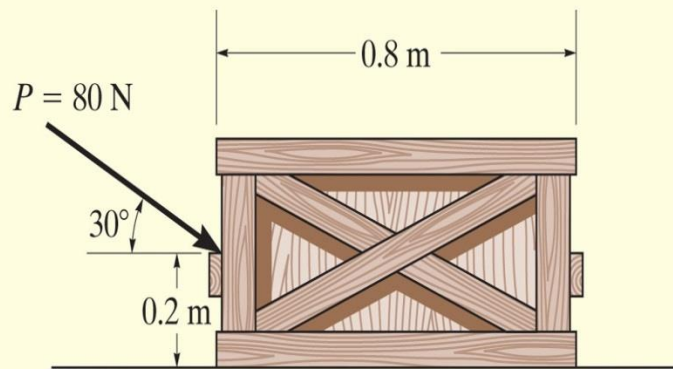
Procedure for analysis

Equations of Equilibrium and Friction.

- Apply the equations of equilibrium and the necessary frictional equations (or conditional equations if tipping is possible) and solve for the unknowns.
- If the problem involves a three-dimensional force system such that it becomes difficult to obtain the force components or the necessary moment arms, apply the equations of equilibrium using Cartesian vectors.

EXAMPLE 8.1

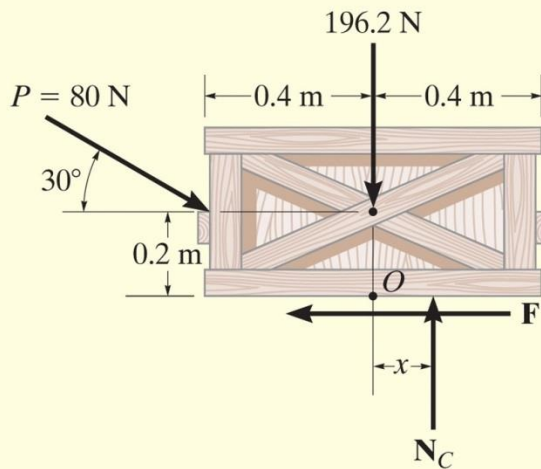
The uniform crate shown in Fig. 8–7*a* has a mass of 20 kg. If a force $P = 80\text{ N}$ is applied to the crate, determine if it remains in equilibrium. The coefficient of static friction is $\mu_s = 0.3$.



(a)

Fig. 8–7

EXAMPLE 8.1 CONTINUED



(b)

SOLUTION

Free-Body Diagram. As shown in Fig. 8-7b, the *resultant* normal force N_C must act a distance x from the crate's center line in order to counteract the tipping effect caused by $\mathbf{P}$. There are *three unknowns*, F , N_C , and x , which can be determined strictly from the *three equations of equilibrium*.

Equations of Equilibrium.

$$\rightarrow \Sigma F_x = 0; \quad 80 \cos 30^\circ \text{ N} - F = 0$$

$$+\uparrow \Sigma F_y = 0; \quad -80 \sin 30^\circ \text{ N} + N_C - 196.2 \text{ N} = 0$$

$$\zeta + \Sigma M_O = 0; \quad 80 \sin 30^\circ \text{ N}(0.4 \text{ m}) - 80 \cos 30^\circ \text{ N}(0.2 \text{ m}) + N_C(x) = 0$$

EXAMPLE 8.1 CONTINUED

Solving,

$$F = 69.3 \text{ N}$$

$$N_C = 236.2 \text{ N}$$

$$x = -0.00908 \text{ m} = -9.08 \text{ mm}$$

Since x is negative it indicates the *resultant* normal force acts (slightly) to the *left* of the crate's center line. No tipping will occur since $x < 0.4 \text{ m}$. Also, the *maximum* frictional force which can be developed at the surface of contact is $F_{\max} = \mu_s N_C = 0.3(236.2 \text{ N}) = 70.9 \text{ N}$. Since $F = 69.3 \text{ N} < 70.9 \text{ N}$, the crate will *not slip*, although it is very close to doing so.

EXAMPLE 8.2

It is observed that when the bed of the dump truck is raised to an angle of $\theta = 25^\circ$ the vending machines will begin to slide off the bed, Fig. 8–8a. Determine the static coefficient of friction between a vending machine and the surface of the truckbed.

SOLUTION

An idealized model of a vending machine resting on the truckbed is shown in Fig. 8–8b. The dimensions have been measured and the center of gravity has been located. We will assume that the vending machine weighs W .



(a)

(© Russell C. Hibbeler)

EXAMPLE

8.2 CONTINUED

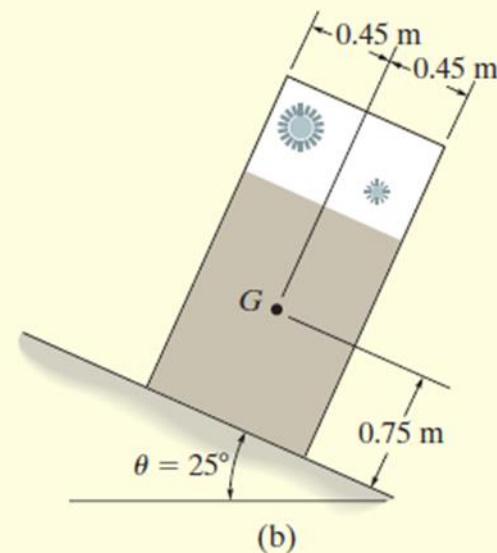
Free-Body Diagram. As shown in Fig. 8–8c, the dimension x is used to locate the position of the resultant normal force $\mathbf{N}$. There are four unknowns, N , F , μ_s , and x .

Equations of Equilibrium.

$$+\searrow \Sigma F_x = 0; \quad W \sin 25^\circ - F = 0 \quad (1)$$

$$+\nearrow \Sigma F_y = 0; \quad N - W \cos 25^\circ = 0 \quad (2)$$

$$\zeta + \Sigma M_O = 0; \quad -(W \sin 25^\circ)(0.75 \text{ m}) + W \cos 25^\circ(x) = 0 \quad (3)$$



EXAMPLE 8.2 CONTINUED

Since slipping impends at $\theta = 25^\circ$, using Eqs. 1 and 2, we have

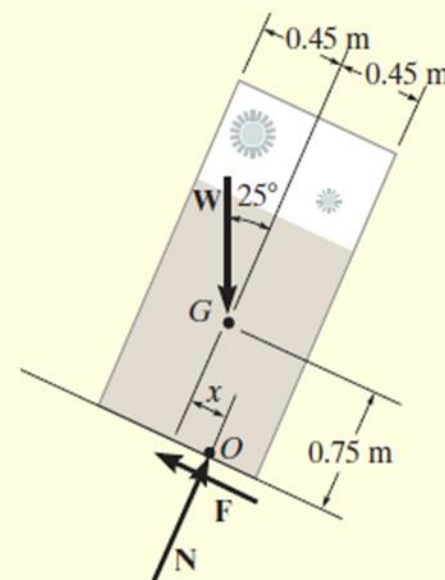
$$F_s = \mu_s N; \quad W \sin 25^\circ = \mu_s (W \cos 25^\circ)$$

$$\mu_s = \tan 25^\circ = 0.466$$

Ans.

The angle of $\theta = 25^\circ$ is referred to as the *angle of repose*, and by comparison, it is equal to the angle of static friction, $\theta = \phi_s$. Notice from the calculation that θ is independent of the weight of the vending machine, and so knowing θ provides a convenient method for determining the coefficient of static friction.

NOTE: From Eq. 3, we find $x = 0.350$ m. Since 0.350 m $<$ 0.45 m, indeed the vending machine will slip before it can tip as observed in Fig. 8–8a.



(c)

Fig. 8–8

EXAMPLE 8.3

The uniform 10-kg ladder in Fig. 8–9*a* rests against the smooth wall at *B*, and the end *A* rests on the rough horizontal plane for which the coefficient of static friction is $\mu_s = 0.3$. Determine the angle of inclination θ of the ladder and the normal reaction at *B* if the ladder is on the verge of slipping.

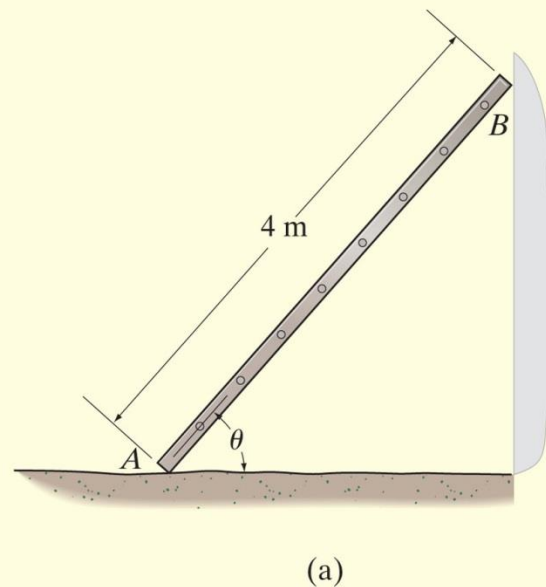
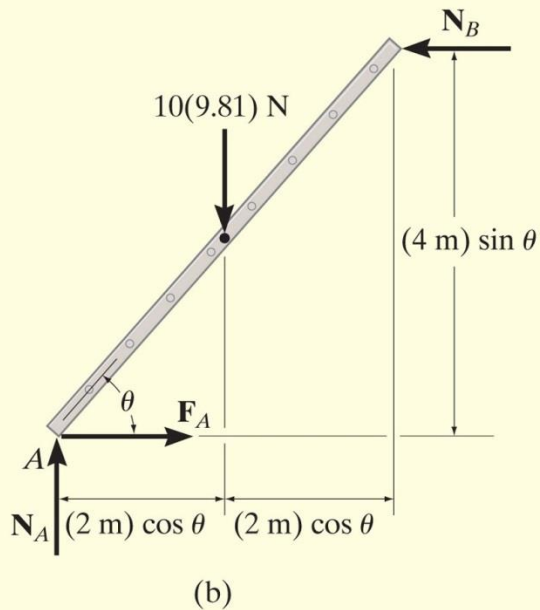


Fig. 8–9

EXAMPLE 8.3 CONTINUED



SOLUTION

Free-Body Diagram. As shown on the free-body diagram, Fig. 8–9b, the frictional force $\mathbf{F}_A$ must act to the right since impending motion at A is to the left.

Equations of Equilibrium and Friction. Since the ladder is on the verge of slipping, then $F_A = \mu_s N_A = 0.3N_A$. By inspection, N_A can be obtained directly.

$$+\uparrow \Sigma F_y = 0; \quad N_A - 10(9.81) \text{ N} = 0 \quad N_A = 98.1 \text{ N}$$

Using this result, $F_A = 0.3(98.1 \text{ N}) = 29.43 \text{ N}$. Now N_B can be found.

$$\pm \Sigma F_x = 0; \quad 29.43 \text{ N} - N_B = 0$$

$$N_B = 29.43 \text{ N} = 29.4 \text{ N} \quad \text{Ans.}$$

Finally, the angle θ can be determined by summing moments about point A.

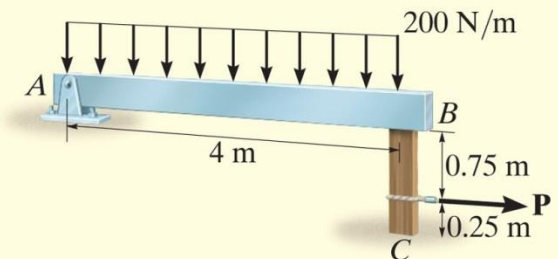
$$\zeta + \Sigma M_A = 0; \quad (29.43 \text{ N})(4 \text{ m}) \sin \theta - [10(9.81) \text{ N}](2 \text{ m}) \cos \theta = 0$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta = 1.6667$$

$$\theta = 59.04^\circ = 59.0^\circ \quad \text{Ans.}$$

EXAMPLE 8.4

Beam AB is subjected to a uniform load of 200 N/m and is supported at B by post BC , Fig. 8–10a. If the coefficients of static friction at B and C are $\mu_B = 0.2$ and $\mu_C = 0.5$, determine the force $\mathbf{P}$ needed to pull the post out from under the beam. Neglect the weight of the members and the thickness of the beam.



(a)

SOLUTION

Free-Body Diagrams. The free-body diagram of the beam is shown in Fig. 8–10b. Applying $\Sigma M_A = 0$, we obtain $N_B = 400 \text{ N}$. This result is shown on the free-body diagram of the post, Fig. 8–10c. Referring to this member, the *four* unknowns F_B , P , F_C , and N_C are determined from the *three* equations of equilibrium and *one* frictional equation applied either at B or C .

Equations of Equilibrium and Friction.

$$\pm \Sigma F_x = 0; \quad P - F_B - F_C = 0 \quad (1)$$

$$+\uparrow \Sigma F_y = 0; \quad N_C - 400 \text{ N} = 0 \quad (2)$$

$$\zeta + \Sigma M_C = 0; \quad -P(0.25 \text{ m}) + F_B(1 \text{ m}) = 0 \quad (3)$$

EXAMPLE 8.4 CONTINUED

(Post Slips at B and Rotates about C.) This requires $F_C \leq \mu_C N_C$ and

$$F_B = \mu_B N_B; \quad F_B = 0.2(400 \text{ N}) = 80 \text{ N}$$

Using this result and solving Eqs. 1 through 3, we obtain

$$P = 320 \text{ N}$$

$$F_C = 240 \text{ N}$$

$$N_C = 400 \text{ N}$$

Since $F_C = 240 \text{ N} > \mu_C N_C = 0.5(400 \text{ N}) = 200 \text{ N}$, slipping at C occurs. Thus the other case of movement must be investigated.

(Post Slips at C and Rotates about B.) Here $F_B \leq \mu_B N_B$ and

$$F_C = \mu_C N_C; \quad F_C = 0.5 N_C \quad (4)$$

Solving Eqs. 1 through 4 yields

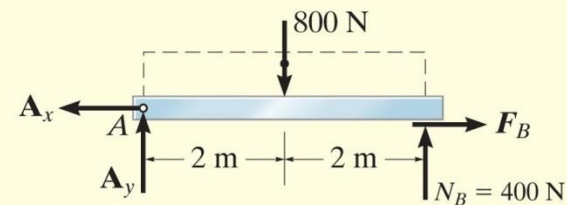
$$P = 267 \text{ N}$$

$$N_C = 400 \text{ N}$$

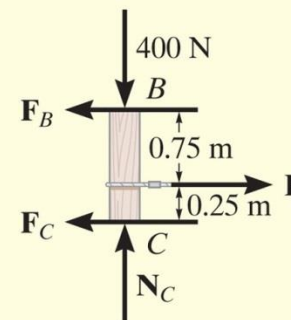
$$F_C = 200 \text{ N}$$

$$F_B = 66.7 \text{ N}$$

Obviously, this case occurs first since it requires a *smaller* value for P .



(b)

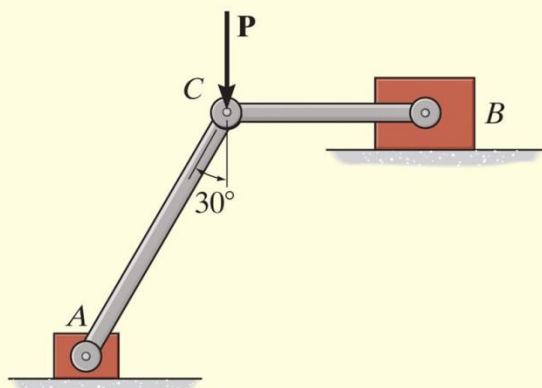


(c)

Fig. 8-10

Ans.

EXAMPLE 8.5



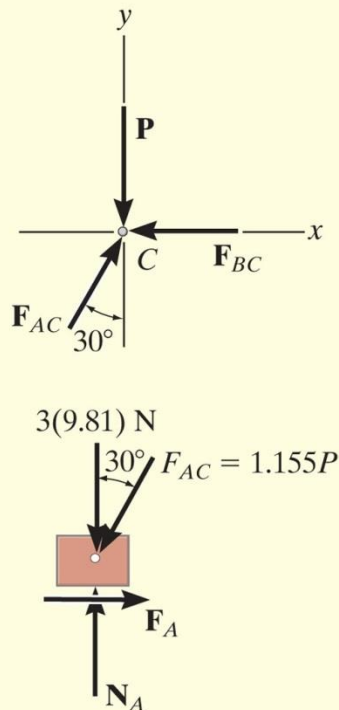
(a)

Blocks A and B have a mass of 3 kg and 9 kg, respectively, and are connected to the weightless links shown in Fig. 8–11a. Determine the largest vertical force $\mathbf{P}$ that can be applied at the pin C without causing any movement. The coefficient of static friction between the blocks and the contacting surfaces is $\mu_s = 0.3$.

SOLUTION

Free-Body Diagram. The links are two-force members and so the free-body diagrams of pin C and blocks A and B are shown in Fig. 8–11b. Since the horizontal component of $\mathbf{F}_{AC}$ tends to move block A to the left, $\mathbf{F}_A$ must act to the right. Similarly, $\mathbf{F}_B$ must act to the left to oppose the tendency of motion of block B to the right, caused by $\mathbf{F}_{BC}$. There are seven unknowns and six available force equilibrium equations, two for the pin and two for each block, so that *only one* frictional equation is needed.

EXAMPLE 8.5 CONTINUED



Equations of Equilibrium and Friction. The force in links AC and BC can be related to P by considering the equilibrium of pin C .

$$\begin{aligned} +\uparrow \Sigma F_y &= 0; & F_{AC} \cos 30^\circ - P &= 0; & F_{AC} &= 1.155P \\ \pm \Sigma F_x &= 0; & 1.155P \sin 30^\circ - F_{BC} &= 0; & F_{BC} &= 0.5774P \end{aligned}$$

Using the result for F_{AC} , for block A ,

$$\pm \Sigma F_x = 0; \quad F_A - 1.155P \sin 30^\circ = 0; \quad F_A = 0.5774P \quad (1)$$

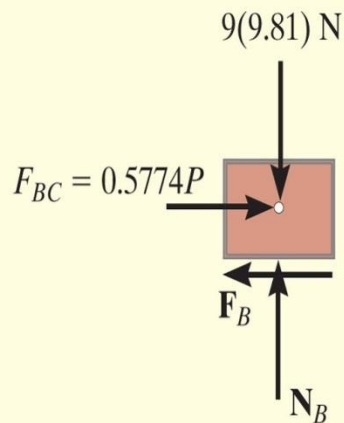
$$\begin{aligned} +\uparrow \Sigma F_y &= 0; \quad N_A - 1.155P \cos 30^\circ - 3(9.81 \text{ N}) = 0; \\ N_A &= P + 29.43 \text{ N} \end{aligned} \quad (2)$$

Using the result for F_{BC} , for block B ,

$$\pm \Sigma F_x = 0; \quad (0.5774P) - F_B = 0; \quad F_B = 0.5774P \quad (3)$$

$$+\uparrow \Sigma F_y = 0; \quad N_B - 9(9.81) \text{ N} = 0; \quad N_B = 88.29 \text{ N}$$

EXAMPLE 8.5 CONTINUED



(b)

Fig. 8-11

Movement of the system may be caused by the initial slipping of *either* block *A* or block *B*. If we assume that block *A* slips first, then

$$F_A = \mu_s N_A = 0.3N_A \quad (4)$$

Substituting Eqs. 1 and 2 into Eq. 4,

$$0.5774P = 0.3(P + 29.43)$$

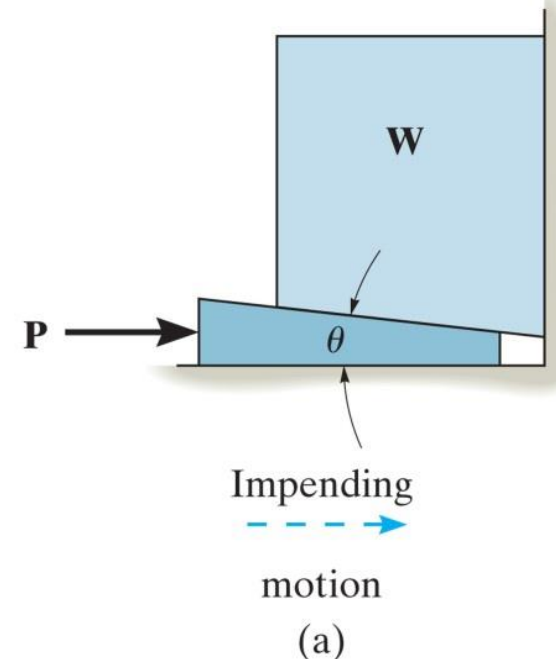
$$P = 31.8 \text{ N}$$

Ans.

Substituting this result into Eq. 3, we obtain $F_B = 18.4 \text{ N}$. Since the maximum static frictional force at *B* is $(F_B)_{\max} = \mu_s N_B = 0.3(88.29 \text{ N}) = 26.5 \text{ N} > F_B$, block *B* will not slip. Thus, the above assumption is correct. Notice that if the inequality were not satisfied, we would have to assume slipping of block *B* and then solve for *P*.

8.3 Wedges

- ✓ A simple machine used to transform an applied force into much larger forces, directed at approximately right angles to the applied force
- ✓ Also used to give slightly move or adjusts to heavy loads
- ✓ Consider the wedge used to lift a block by applying a force **P** to the wedge

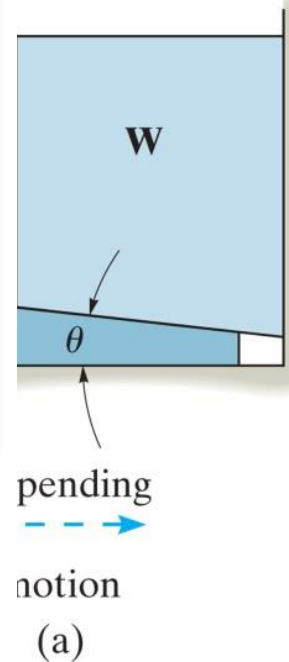


8.3 Wedges



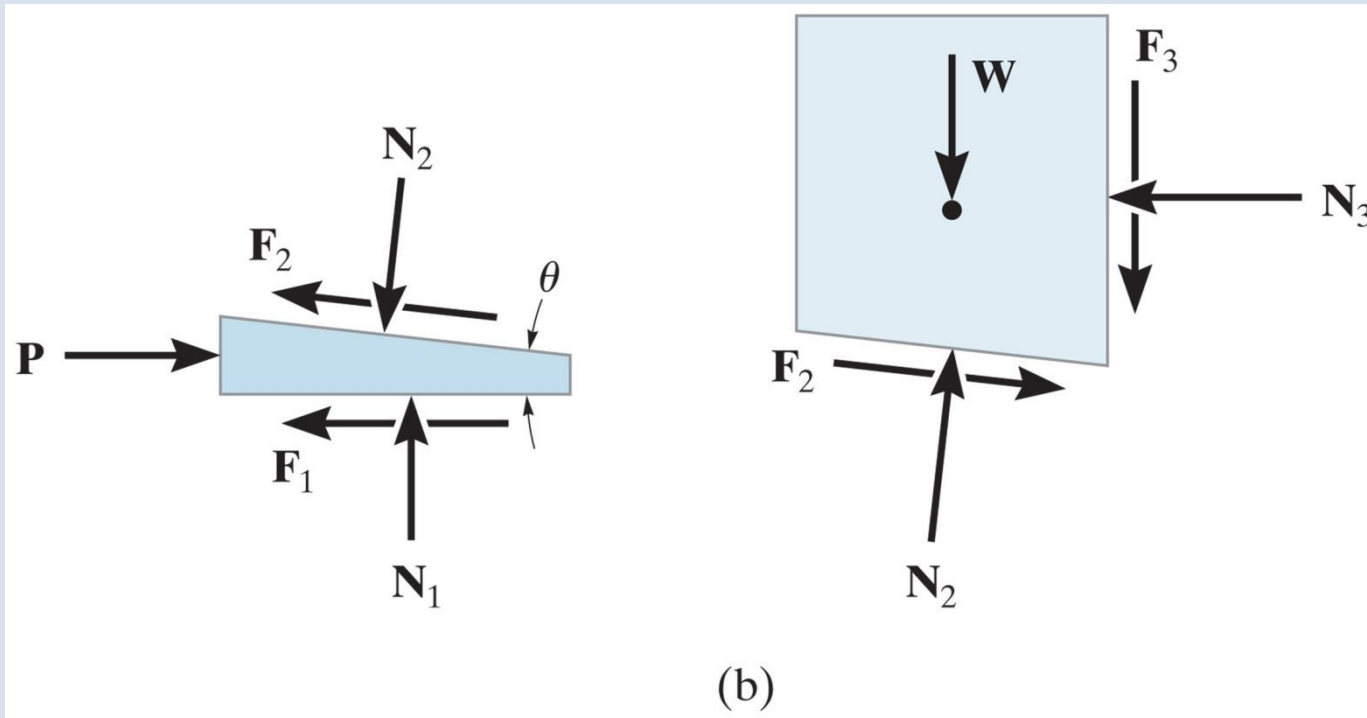
Figure: 08_PH005

Wedges are often used to adjust the elevation of structural or mechanical parts. Also, they provide stability for objects such as this pipe.



8.3 Wedges

- ✓ FBD of the block and the wedge



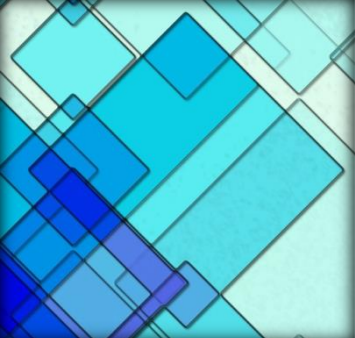
- ✓ Exclude the weight of the wedge since it is small compared to weight of the weight W of the block

8.3 Wedges

- ✓ Frictional forces $\mathbf{F}_1$ and $\mathbf{F}_2$ must oppose the motion of the wedge
- ✓ Frictional force $\mathbf{F}_3$ of the wall on the block must act downward so as to oppose the block's upward motion
- ✓ The Locations of the resultant normal forces are not important since neither the block or the wedge will "tip"
- ✓ Moment equilibrium equations will not be considered
- ✓ 7 unknowns – force P and 6 normal and frictional force

8.3 Wedges

- ✓ 7 available equations consist of 4 force equilibrium equations, $\Sigma F_x = 0$, $\Sigma F_y = 0$ applied to the wedge and block and three frictional equation, $F = \mu N$, applied at each surface of contact
- ✓ If the block is to be **lowered**, the frictional forces will act in a sense opposite that shown in Fig 8-12b
- ✓ Applied force **P** will act to the right if the coefficient of friction is small or the wedge angle θ is large



8.3 Wedges

- ✓ Otherwise, **P** may have a reverse sense of direction in order to **pull** on the wedge to remove it
- ✓ If **P** is not applied and friction forces hold the block in place, then the wedge is referred to as self-locking

EXAMPLE 8.6

The uniform stone in Fig. 8–13*a* has a mass of 500 kg and is held in the horizontal position using a wedge at *B*. If the coefficient of static friction is $\mu_s = 0.3$ at the surfaces of contact, determine the minimum force *P* needed to remove the wedge. Assume that the stone does not slip at *A*.

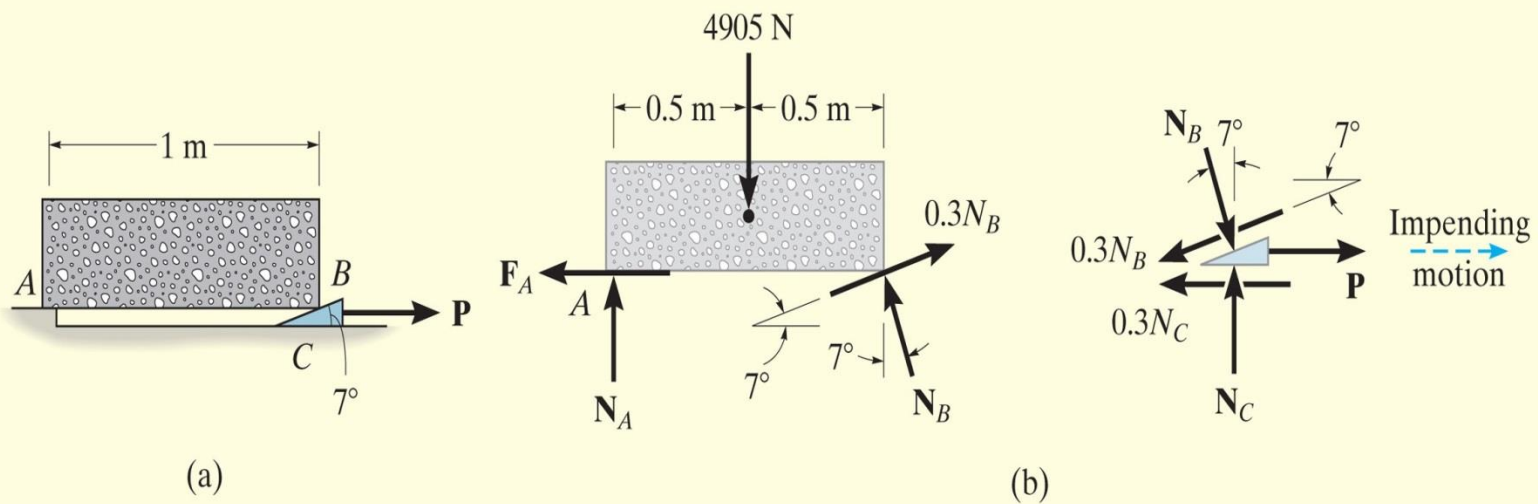


Fig. 8–13

EXAMPLE 8.6 CONTINUED

SOLUTION

The minimum force P requires $F = \mu_s N$ at the surfaces of contact with the wedge. The free-body diagrams of the stone and wedge are shown in Fig. 8–13*b*. On the wedge the friction force opposes the impending motion, and on the stone at A , $F_A \leq \mu_s N_A$, since slipping does not occur there. There are five unknowns. Three equilibrium equations for the stone and two for the wedge are available for solution. From the free-body diagram of the stone,

$$\begin{aligned}\zeta + \Sigma M_A = 0; \quad & -4905 \text{ N}(0.5 \text{ m}) + (N_B \cos 7^\circ \text{ N})(1 \text{ m}) \\ & + (0.3N_B \sin 7^\circ \text{ N})(1 \text{ m}) = 0 \\ N_B = & 2383.1 \text{ N}\end{aligned}$$

EXAMPLE 8.6 CONTINUED

Using this result for the wedge, we have

$$+\uparrow \Sigma F_y = 0; \quad N_C - 2383.1 \cos 7^\circ \text{ N} - 0.3(2383.1 \sin 7^\circ \text{ N}) = 0$$

$$N_C = 2452.5 \text{ N}$$

$$\pm \Sigma F_x = 0; \quad 2383.1 \sin 7^\circ \text{ N} - 0.3(2383.1 \cos 7^\circ \text{ N}) +$$

$$P - 0.3(2452.5 \text{ N}) = 0$$

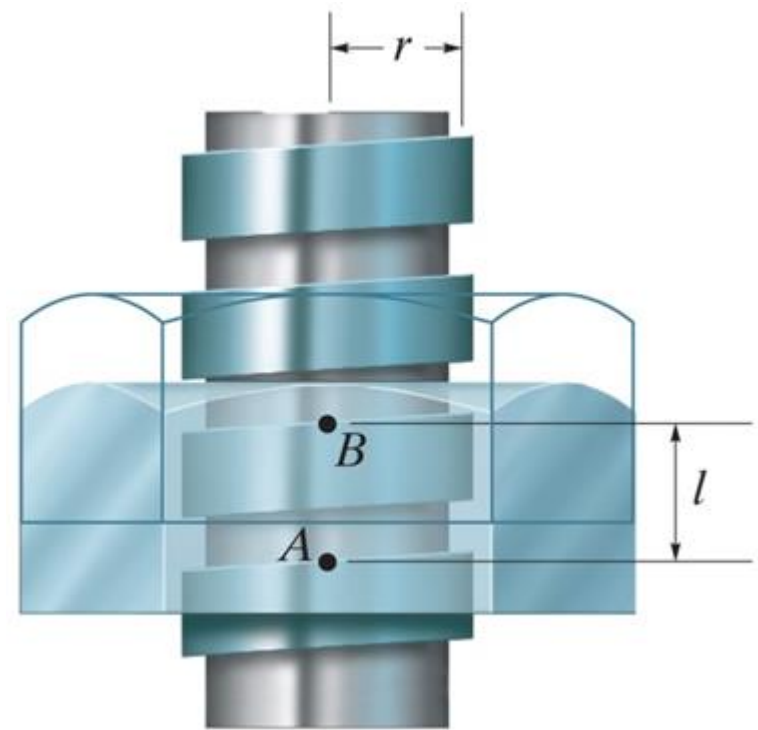
$$P = 1154.9 \text{ N} = 1.15 \text{ kN} \quad \text{Ans.}$$

NOTE: Since P is positive, indeed the wedge must be pulled out. If P were zero, the wedge would remain in place (self-locking) and the frictional forces developed at B and C would satisfy $F_B < \mu_s N_B$ and $F_C < \mu_s N_C$.

8.4 Frictional Forces on Screws

Screws used as fasteners; sometimes used to transmit power or motion from one part of the machine to another.

Square-threaded screw: thought as an inclined plane or wedge wrapped around a cylinder



8.4 Friction

Screws used as fasteners are sometimes used to transmit power or motion from one part of the machine to another.

Square-threaded screws are often thought of as an inclined plane or wedge wrapped around a cylinder.

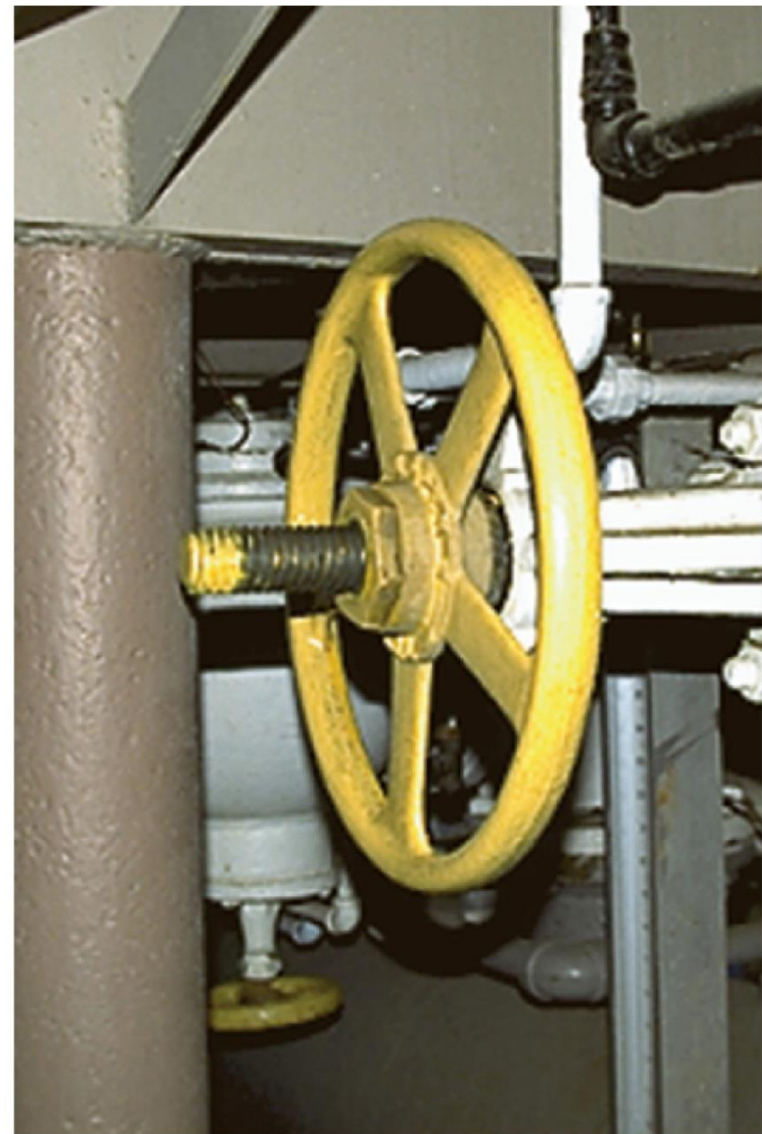


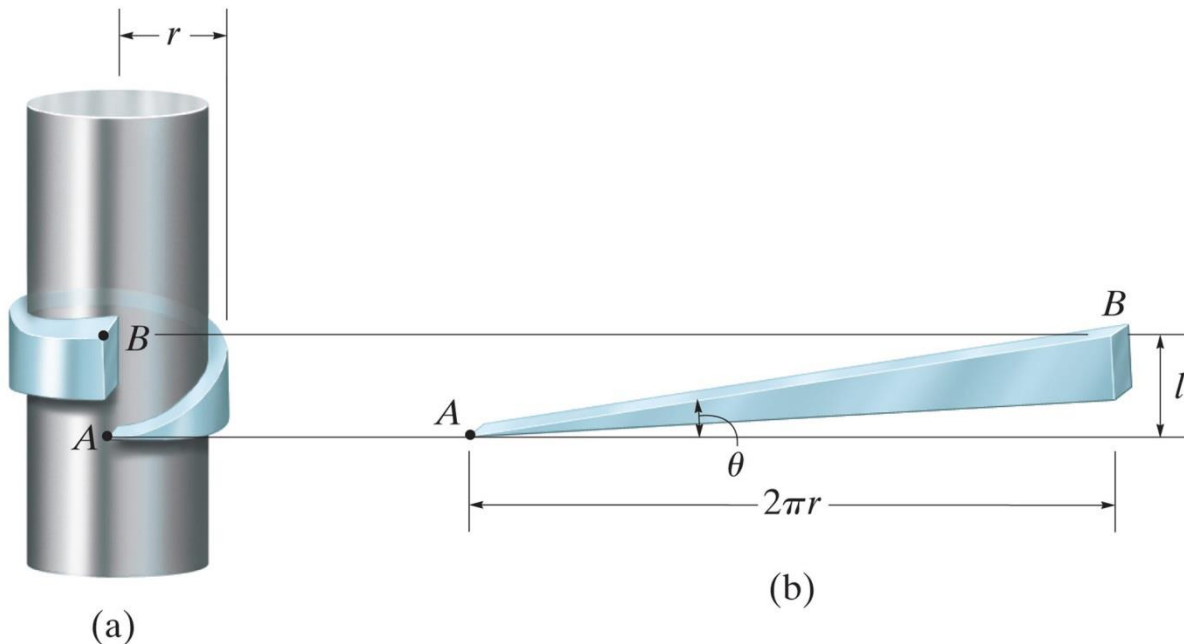
Figure: 08_PH006

Square-threaded screws find applications on valves, jacks, and vises, where particularly large forces must be developed along the axis of the screw.

8.4 Frictional Forces on Screws

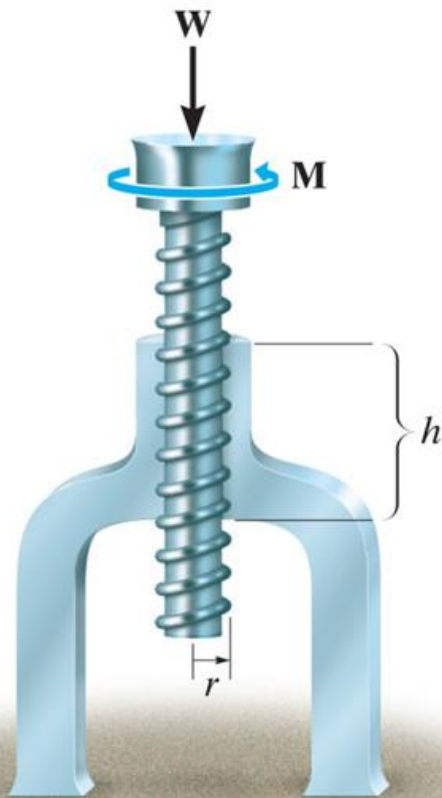
If unwind the thread by one revolution $\rightarrow$

$$\theta = \tan^{-1} (l / 2\pi r)$$

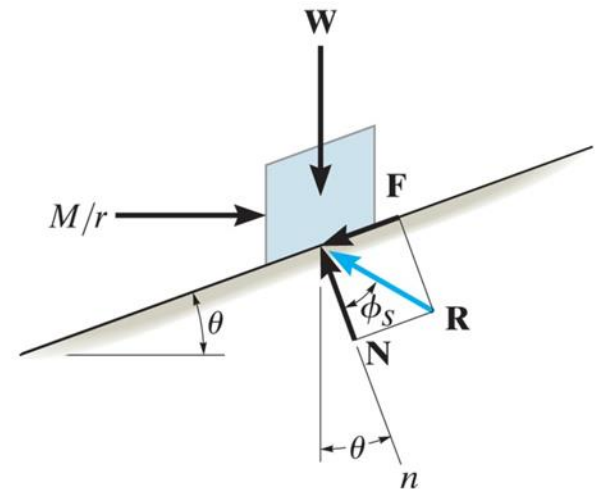


8.4 Frictional Forces on Screws

Upward Impending Motion



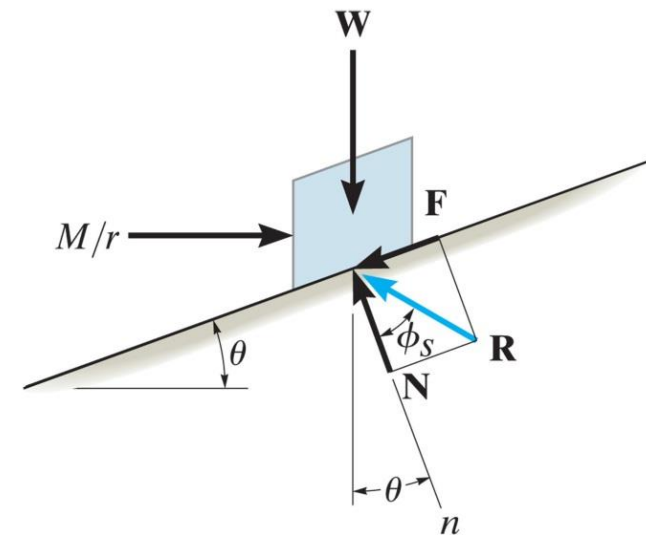
A square-threaded screw is subjected to upward impending motion caused by the applied torsional moment M



Upward screw motion
(a)

8.4 Frictional Forces on Screws

- ✓ The force W is the vertical force acting on the thread or the axial force applied to the shaft,
- ✓ M / r is the resultant horizontal force produced by the couple moment M about the axis of the shaft.
- ✓ The reaction R of the groove on the thread has both frictional and normal components, where $F = \mu_s N$.



Upward screw motion
(a)

8.4 Frictional Forces on Screws

- ✓ The angle of static friction is $\phi_s = \tan^{-1} (F/N) = \tan^{-1} \mu_s$
- ✓ Applying the force equations of equilibrium along the horizontal and vertical axes, we have

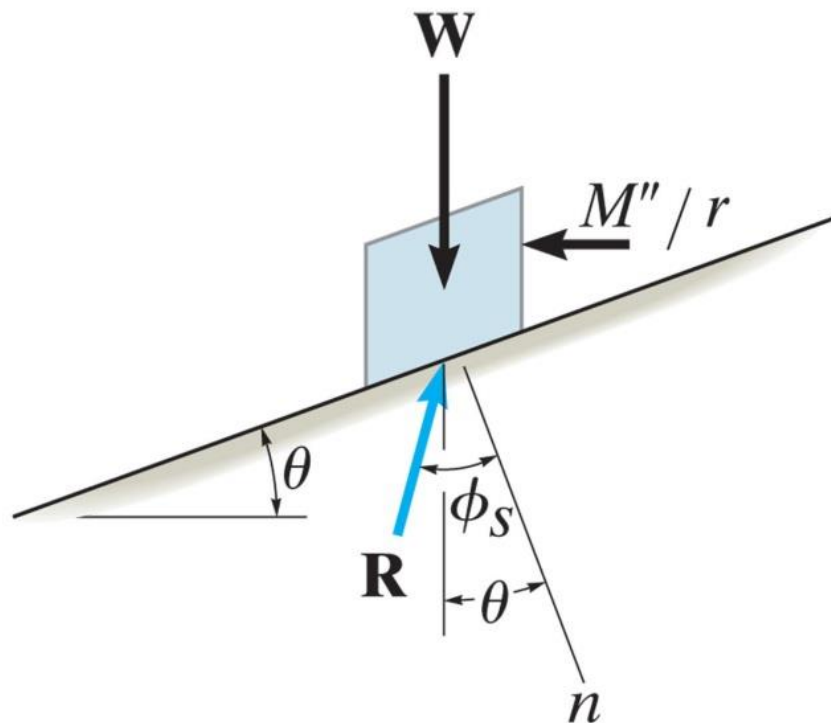
$$\begin{aligned} + \rightarrow \sum F_x &= 0; & M/r - R \sin(\theta + \phi_s) &= 0 \\ + \uparrow \sum F_y &= 0; & R \cos(\theta + \phi_s) - W &= 0 \end{aligned}$$

- ✓ Eliminating R from these equations, we obtain

$$M = rW \tan(\theta + \phi_s)$$

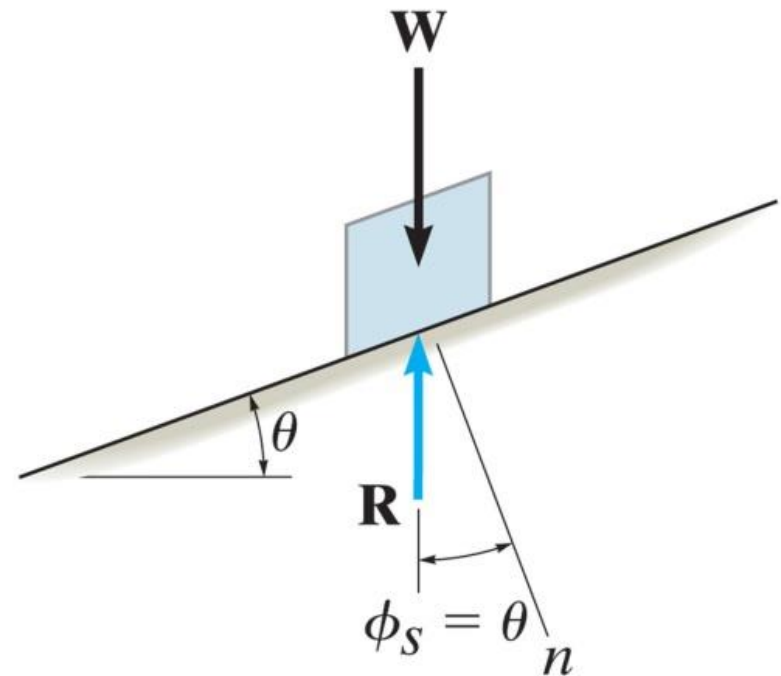
8.4 Frictional Forces on Screws

Self-Locking Screw



Downward screw motion ($\theta < \phi_s$)

(d)



Self-locking screw ($\theta = \phi_s$)
(on the verge of rotating downward)

(b)

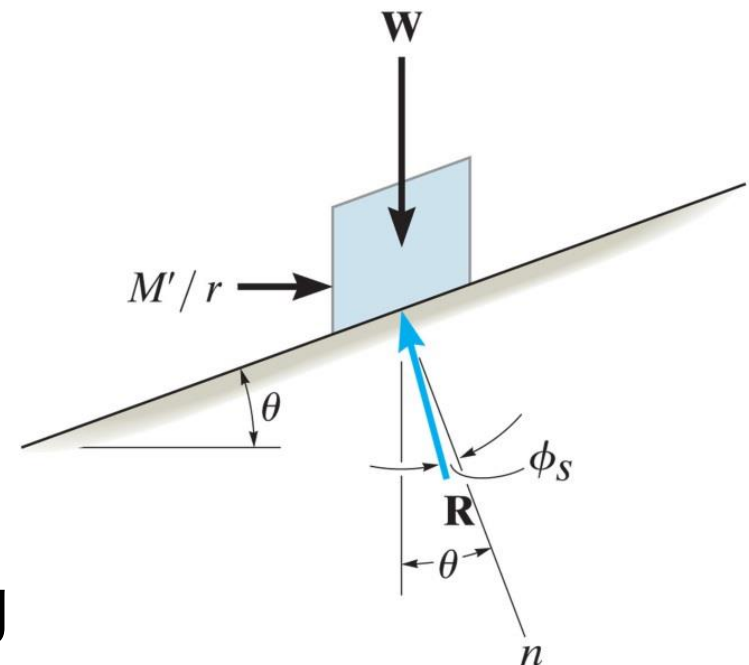


8.4 Frictional Forces on Screws

Downward Impending Motion

$$(\theta > \phi_s)$$

- ✓ If the screw is **not** self-locking
→ a moment M' to prevent the screw from winding downward.
- ✓ A horizontal force M' / r is required to push against the thread to prevent it from sliding down the plane.



Downward screw motion ($\theta > \phi_s$)

(c)

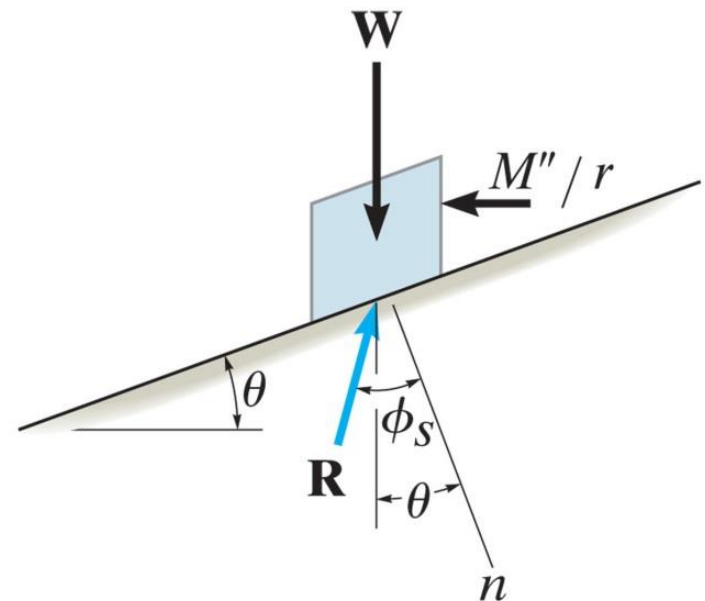
$$M' = rW \tan(\theta - \phi_s)$$

8.4 Frictional Forces on Screws

Downward Impending Motion

$$(\phi_s > \theta)$$

- ✓ If a screw is self-locking, a couple moment M'' must be applied to the screw in the **opposite** direction to wind the screw downward ($\phi_s > \theta$).
- ✓ This causes a **reverse** horizontal force M''/r that pushes the thread down.



Downward screw motion ($\theta < \phi_s$)

(d)

$$M'' = rW \tan(\phi_s - \theta)$$

EXAMPLE 8.7

The turnbuckle shown in Fig. 8–17 has a square thread with a mean radius of 5 mm and a lead of 2 mm. If the coefficient of static friction between the screw and the turnbuckle is $\mu_s = 0.25$, determine the moment $\mathbf{M}$ that must be applied to draw the end screws closer together.



(© Russell C. Hibbeler)

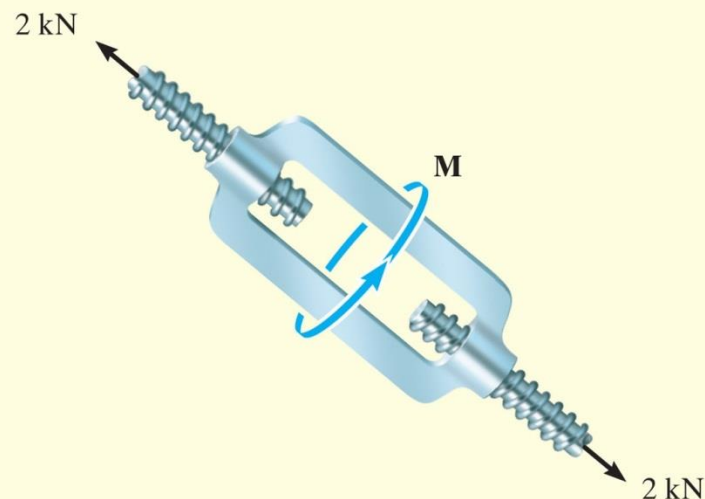


Fig. 8–17

EXAMPLE 8.7 CONTINUED

SOLUTION

The moment can be obtained by applying Eq. 8–3. Since friction at *two screws* must be overcome, this requires

$$M = 2[rW \tan(\theta + \phi_s)] \quad (1)$$

Here $W = 2000 \text{ N}$, $\phi_s = \tan^{-1} \mu_s = \tan^{-1}(0.25) = 14.04^\circ$, $r = 5 \text{ mm}$, and $\theta = \tan^{-1}(l/2\pi r) = \tan^{-1}(2 \text{ mm}/[2\pi(5 \text{ mm})]) = 3.64^\circ$. Substituting these values into Eq. 1 and solving gives

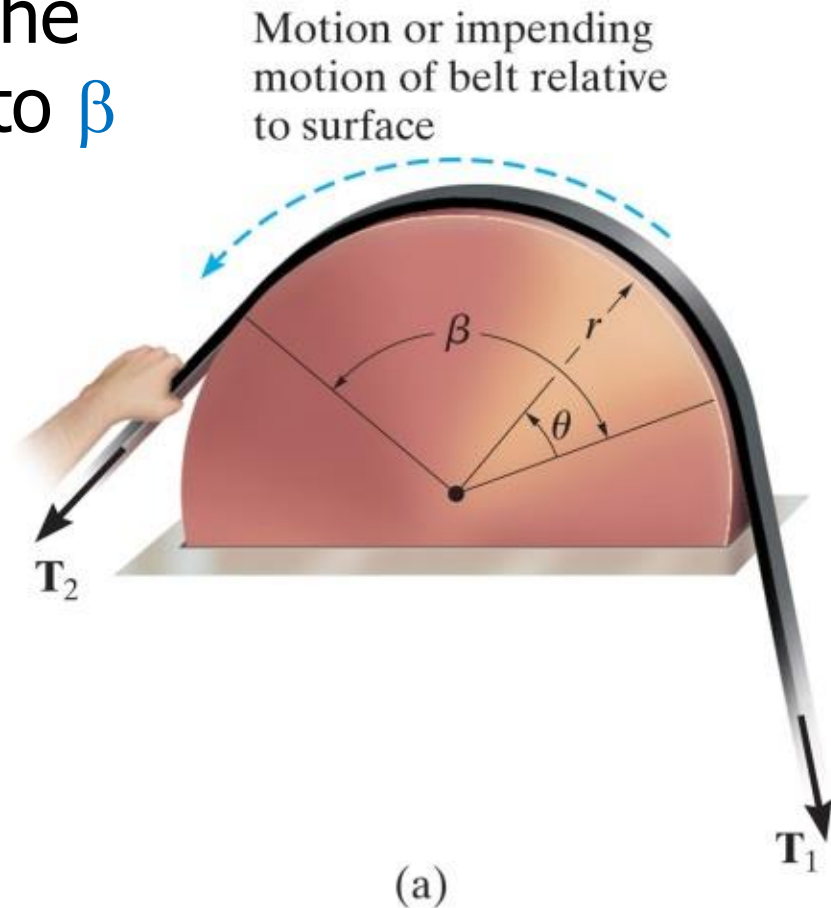
$$M = 2[(2000 \text{ N})(5 \text{ mm}) \tan(14.04^\circ + 3.64^\circ)]$$

$$= 6374.7 \text{ N} \cdot \text{mm} = 6.37 \text{ N} \cdot \text{m} \quad \text{Ans.}$$

NOTE: When the moment is *removed*, the turnbuckle will be self-locking; i.e., it will not unscrew since $\phi_s > \theta$.

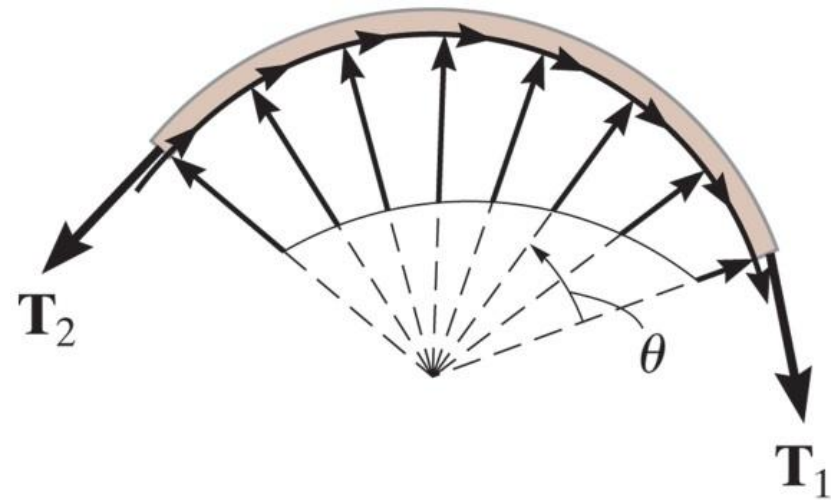
8.5 Frictional Forces on Flat Belts

- ✓ Consider a flat belt passing over a fixed curved surface with the total angle of contact equal to β radians.
- ✓ μ : coefficient of friction
- ✓ $T_2 > T_1$



8.5 Frictional Forces on Flat Belts

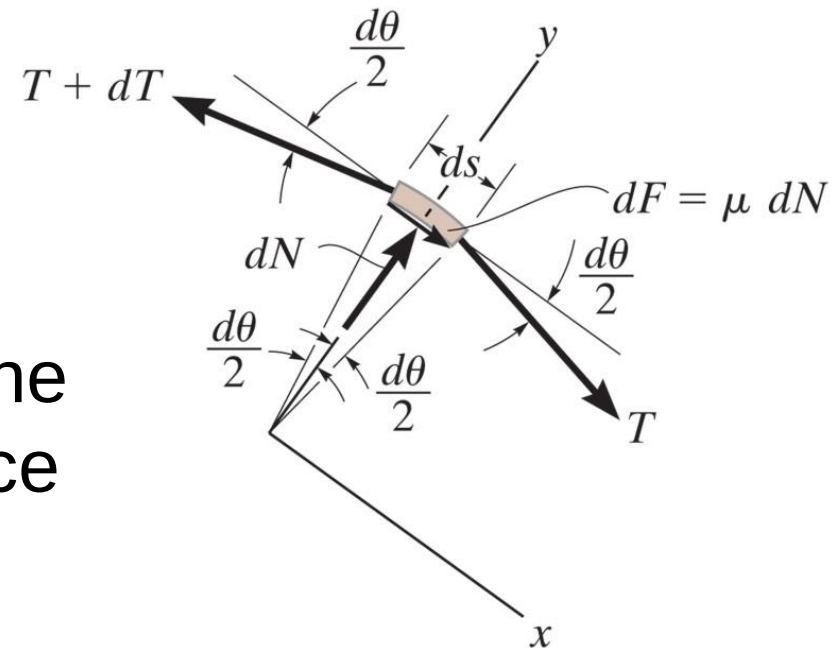
- ✓ Normal and Frictional forces will **vary** both in magnitude and direction.
- ✓ Due to **unknown** distribution, the analysis of the problem will first require a study of the forces acting on a differential element of the belt.



(b)

8.5 Frictional Forces on Flat Belts

- ✓ Consider FBD of an element having a length ds
- ✓ Assuming either impending motion or motion of the belt, the magnitude of the frictional force $dF = \mu dN$



$$\sum F_x = 0; \quad T \cos\left(\frac{d\theta}{2}\right) + \mu dN - (T + dT) \cos\left(\frac{d\theta}{2}\right) = 0$$

$$\sum F_y = 0; \quad dN - (T + dT) \sin\left(\frac{d\theta}{2}\right) - T \sin\left(\frac{d\theta}{2}\right) = 0$$

(c)

8.5 Frictional Forces on Flat Belts

We have

$$\mu dN = dT$$

$$dN = Td\theta$$

Eliminating dN

$$\frac{dT}{T} = \mu d\theta$$

Integrating the equation

$$T = T_1, \theta = 0, T = T_2, \theta = \beta$$

$$\int_{T_1}^{T_2} \frac{dT}{T} = \mu \int_0^\beta d\theta$$

$$\ln \frac{T_2}{T_1} = \mu\beta$$

Solving for T_2

$$T_2 = T_1 e^{\mu\beta}$$

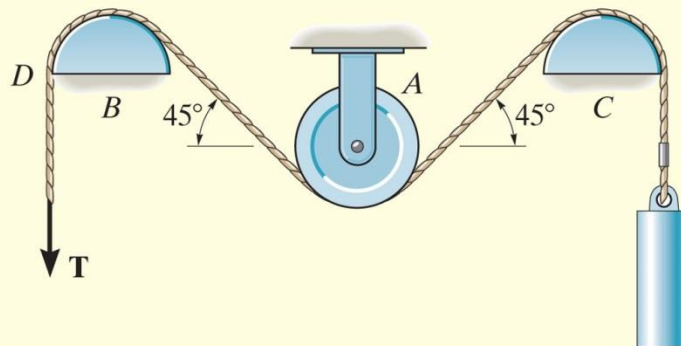


Figure: 08_PH007

Flat or V-belts are often used to transmit the torque developed by a motor to a wheel attached to a pump, fan or blower.

EXAMPLE 8.8

The maximum tension that can be developed in the cord shown in Fig. 8–19a is 500 N. If the pulley at A is free to rotate and the coefficient of static friction at the fixed drums B and C is $\mu_s = 0.25$, determine the largest mass of the cylinder that can be lifted by the cord.



(a)

EXAMPLE 8.8 CONTINUED

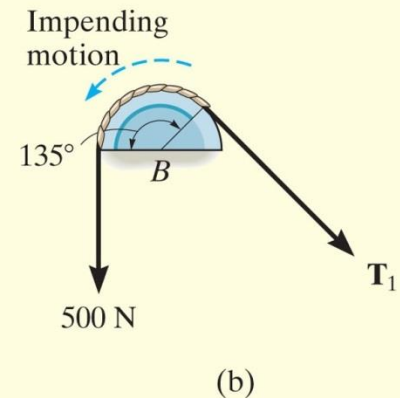
SOLUTION

Lifting the cylinder, which has a weight $W = mg$, causes the cord to move counterclockwise over the drums at B and C ; hence, the maximum tension T_2 in the cord occurs at D . Thus, $F = T_2 = 500$ N. A section of the cord passing over the drum at B is shown in Fig. 8-19b. Since $180^\circ = \pi$ rad the angle of contact between the drum and the cord is $\beta = (135^\circ/180^\circ)\pi = 3\pi/4$ rad. Using Eq. 8-6, we have

$$T_2 = T_1 e^{\mu_s \beta}; \quad 500 \text{ N} = T_1 e^{0.25[(3/4)\pi]}$$

Hence,

$$T_1 = \frac{500 \text{ N}}{e^{0.25[(3/4)\pi]}} = \frac{500 \text{ N}}{1.80} = 277.4 \text{ N}$$



EXAMPLE 8.8 CONTINUED

Since the pulley at A is free to rotate, equilibrium requires that the tension in the cord remains the *same* on both sides of the pulley.

The section of the cord passing over the drum at C is shown in Fig. 8–19c. The weight $W < 277.4$ N. Why? Applying Eq. 8–6, we obtain

$$T_2 = T_1 e^{\mu_s \beta}; \quad 277.4 \text{ N} = W e^{0.25[(3/4)\pi]}$$

$$W = 153.9 \text{ N}$$

so that

$$\begin{aligned} m &= \frac{W}{g} = \frac{153.9 \text{ N}}{9.81 \text{ m/s}^2} \\ &= 15.7 \text{ kg} \end{aligned}$$

Ans.

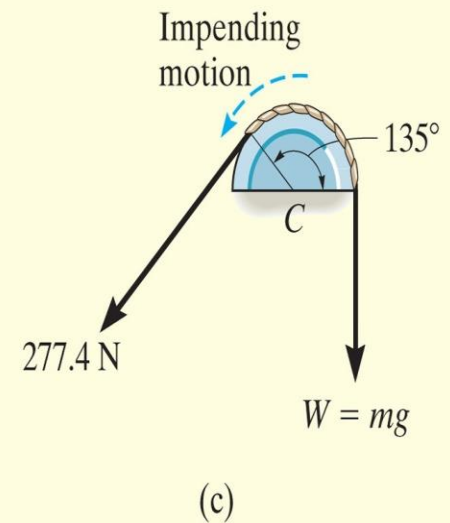
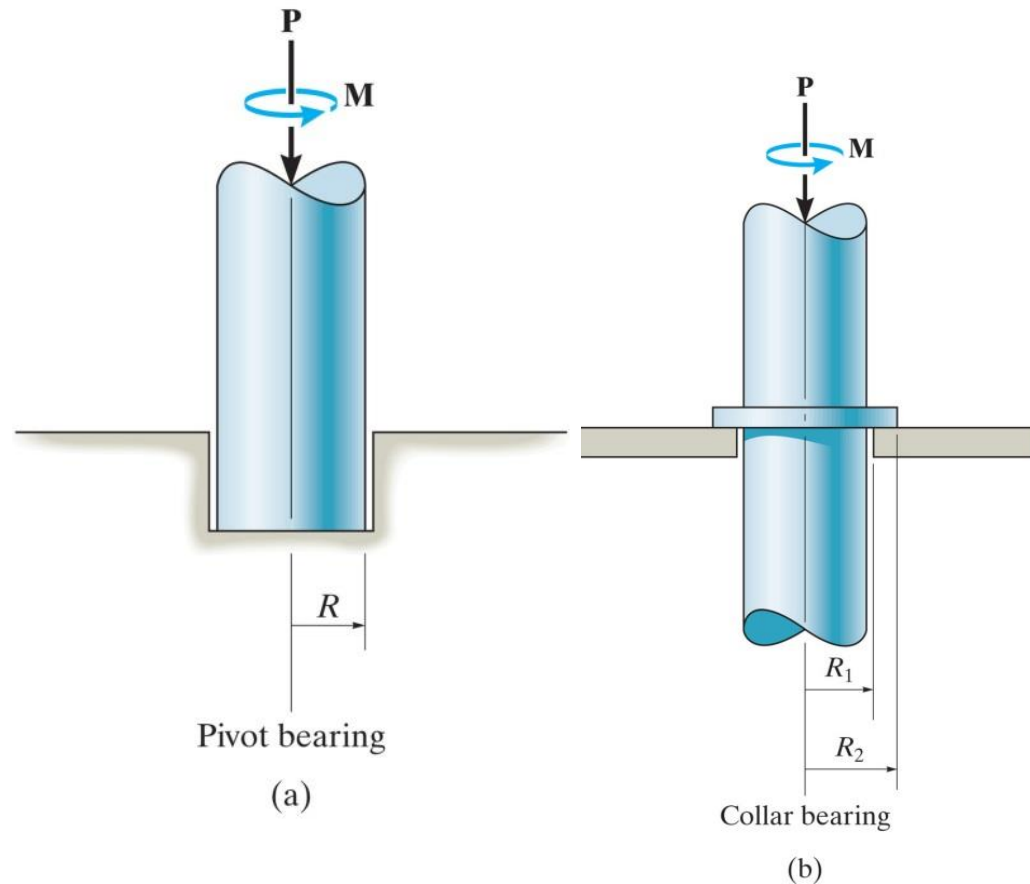


Fig. 8–19

8.6 Frictional Forces on Collar Bearings, Pivot Bearings and Disks

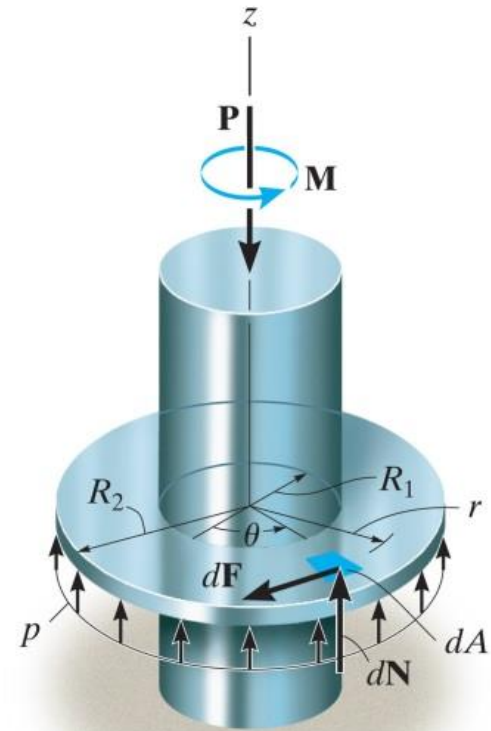
- ✓ Pivot and collar bearings are used to support axial load on a rotating shaft
- ✓ Laws of dry friction is applied to determine the moment M needed to turn the shaft when it supports an axial force P



8.6 Frictional Forces on Collar Bearings, Pivot Bearings and Disks

Frictional Analysis

- ✓ The collar bearing on the shaft is subjected to an axial force $\mathbf{P}$ and has a total contact area $\pi(R_2^2 - R_1^2)$
- ✓ Normal pressure p is considered to be **uniformly distributed** over this area
- ✓ Since $\sum F_z = 0$, p measured as a force per unit area $p = P / \pi(R_2^2 - R_1^2)$



8.6 Frictional Forces on Collar Bearings, Pivot Bearings and Disks

- ✓ The moment needed to cause impending rotation of the shaft can be determined from moment equilibrium about the **z axis**
- ✓ A differential area element **$dA = (r \, d\theta)(dr)$** is subjected to both a normal force **$dN = p \, dA$** , and an associated frictional force,

$$dF = \mu_s dN = \mu_s p \, dA = \frac{\mu_s P}{\pi(R_2^2 - R_1^2)} dA$$

8.6 Frictional Forces on Collar Bearings, Pivot Bearings and Disks

- ✓ Normal force does not create a moment about z axis of the shaft, however, frictional force does, $dM = r dF$
- ✓ For impending rotational motion

$$\Sigma M_z = 0;$$

$$M - \int_A r dF = 0$$

8.6 Frictional Forces on Collar Bearings, Pivot Bearings and Disks

✓ Integrating over entire bearing area

$$M = \int_{R_1}^{R_2} \int_0^{2\pi} r \left[\frac{\mu_s P}{\pi(R_2^2 - R_1^2)} \right] (r d\theta dr) = \frac{\mu_s P}{\pi(R_2^2 - R_1^2)} \int_{R_1}^{R_2} r^2 dr \int_0^{2\pi} d\theta$$

Or

$$M = \frac{2}{3} \mu_s P \left(\frac{R_2^3 - R_1^3}{R_2^2 - R_1^2} \right)$$

8.6 Frictional Forces on Collar Bearings, Pivot Bearings and Disks

- ✓ Frictional moment developed at the end of the shaft, when it is rotating at constant speed, can be found by substituting μ_k for μ_s
- ✓ In Fig.8-20a, when $R_2 = R$ and $R_1 = 0$,
→ $M = \frac{2}{3} \mu_s PR$
- ✓ Eqs. 8-7 and 8-8 apply only to bearings subjected to constant pressure
- ✓ If the pressure is not uniform, a variation of the pressure as a function of the bearing area must be determined to obtain moment.

8.6 Frictional Forces on Collar Bearings, Shafts, and Collar Bearings



Figure: 08_PH008

The motor that turns the disk of this sanding machine develops a torque that must overcome the frictional forces acting on the disk.

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speed, can

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igs

tion of the
area must

EXAMPLE 8.9

The uniform bar shown in Fig. 8–22*a* has a weight of 20 N. If it is assumed that the normal pressure acting at the contacting surface varies linearly along the length of the bar as shown, determine the couple moment $\mathbf{M}$ required to rotate the bar. Assume that the bar's width is negligible in comparison to its length. The coefficient of static friction is equal to $\mu_s = 0.3$.

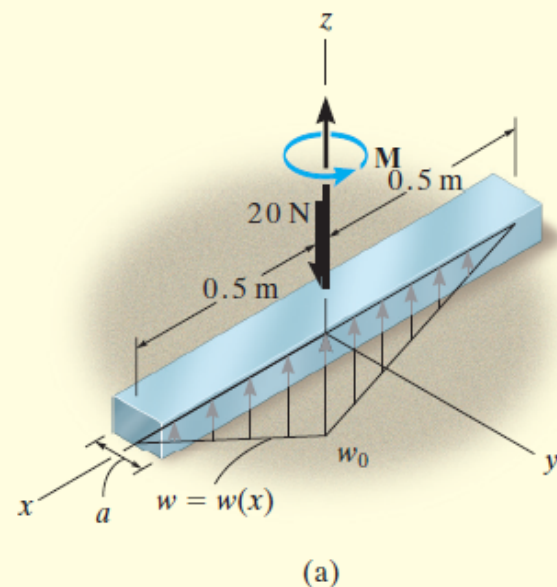
SOLUTION

A free-body diagram of the bar is shown in Fig. 8–22*b*. The intensity w_0 of the distributed load at the center ($x = 0$) is determined from vertical force equilibrium, Fig. 8–22*a*.

$$+\uparrow \Sigma F_z = 0; \quad -20 \text{ N} + 2 \left[\frac{1}{2} (0.5 \text{ m}) w_0 \right] = 0 \quad w_0 = 40 \text{ N/m}$$

Since $w = 0$ at $x = 0.5 \text{ m}$, the distributed load expressed as a function of x is

$$w = (40 \text{ N/m}) \left(1 - \frac{x}{0.5 \text{ m}} \right) = 40 - 80x$$



EXAMPLE 8.9 CONTINUED

The magnitude of the normal force acting on a differential segment of area having a length dx is therefore

$$dN = w dx = (40 - 80x)dx$$

The magnitude of the frictional force acting on the same element of area is

$$dF = \mu_s dN = 0.3(40 - 80x)dx$$

Hence, the moment created by this force about the z axis is

$$dM = x dF = 0.3(40x - 80x^2)dx$$

The summation of moments about the z axis of the bar is determined by integration, which yields

$$\Sigma M_z = 0; \quad M - 2 \int_0^{0.5 \text{ m}} (0.3)(40x - 80x^2) dx = 0$$

$$M = 6 \left(x^2 - \frac{4x^3}{3} \right) \Big|_0^{0.5 \text{ m}}$$

$$M = 0.5 \text{ N} \cdot \text{m}$$

$$M = 24 \left(\frac{x^2}{2} - \frac{2x^3}{3} \right) \Big|_0^{0.5 \text{ m}}$$

$$= 1.00 \text{ N} \cdot \text{m}$$

Ans.

Ans.

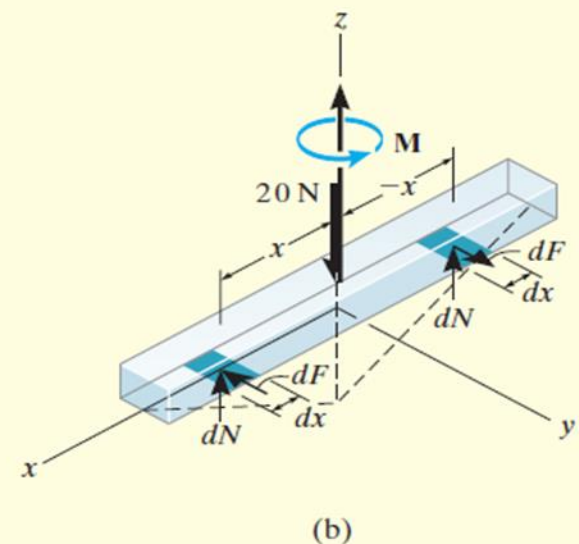
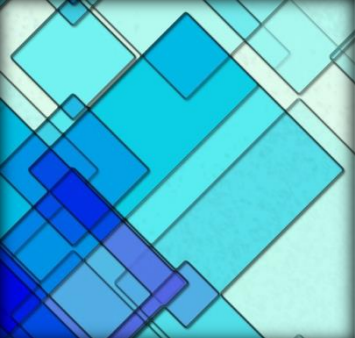


Fig. 8-22



8.7 Frictional Forces on Journal Bearings

- ✓ When a shaft or axle is subjected to lateral loads, a **journal bearing** is used for support
- ✓ Provided the bearing is not / partially lubricated, analysis of the frictional resistance can be based on the **laws of dry friction**



Figure: 08_PH009

Unwinding the cable from this spool requires overcoming friction from the supporting shaft.

s on Journal

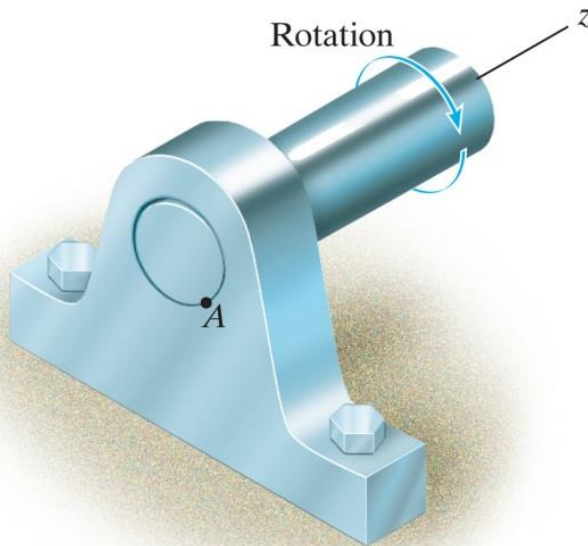
d to lateral
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tially
nal resistance
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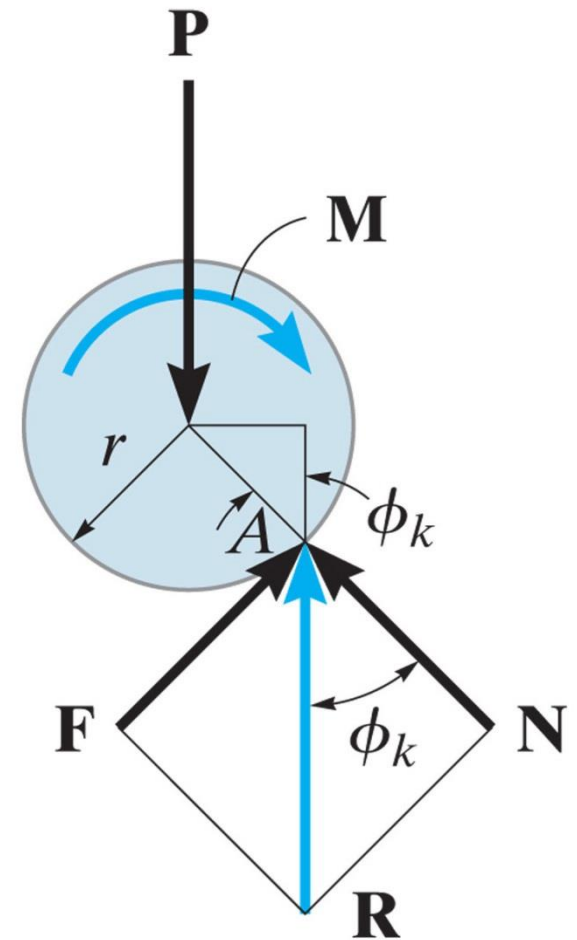
8.7 Frictional Forces on Journal Bearings

Frictional Analysis

- ✓ If the vertical load is $\mathbf{P}$, the bearing reactive force $\mathbf{R}$ acting at A is equal and opposite to $\mathbf{P}$



(a)



(b)

8.7 Frictional Forces on Journal Bearings

- ✓ The moment needed to maintain constant rotation of the shaft can be found by the summation of moments about the z axis of the shaft,

$$\Sigma M_z = 0; \quad M - (R \sin \phi_k) r = 0$$

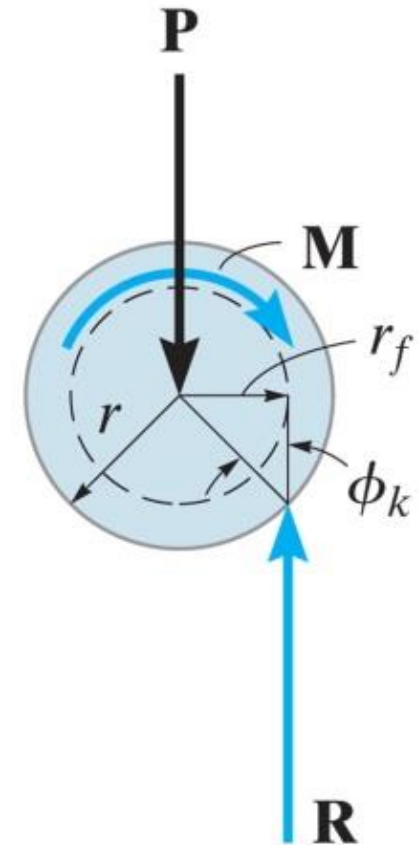
Or

$$M = Rr \sin \phi_k$$

8.7 Frictional Forces on Journal Bearings

- ✓ In fig2-23C, $r \sin \phi_k = r_f$.
- ✓ The dashed circle with radius r_f is called the **friction circle**.
- ✓ If the bearing is partially lubricated, μ_k is small,
→ $\sin \phi_k \approx \tan \phi_k \approx \mu_k$
- ✓ Frictional resistance

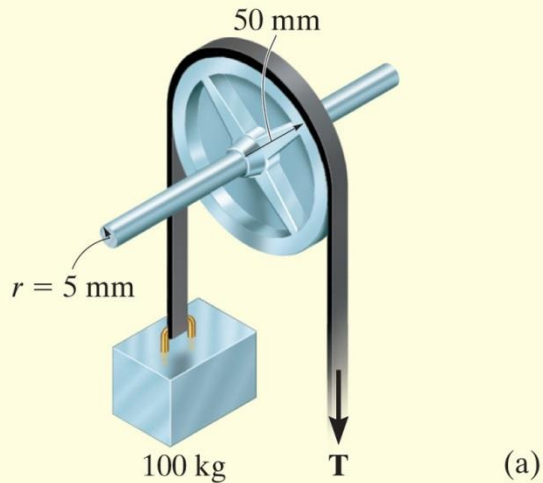
$$M \approx Rr\mu_k$$



(c)

8.10

The 100-mm-diameter pulley shown in Fig. 8–24*a* fits loosely on a 10-mm-diameter shaft for which the coefficient of static friction is $\mu_s = 0.4$. Determine the minimum tension T in the belt needed to (a) raise the 100-kg block and (b) lower the block. Assume that no slipping occurs between the belt and pulley and neglect the weight of the pulley.



EXAMPLE 8.10 CONTINUED

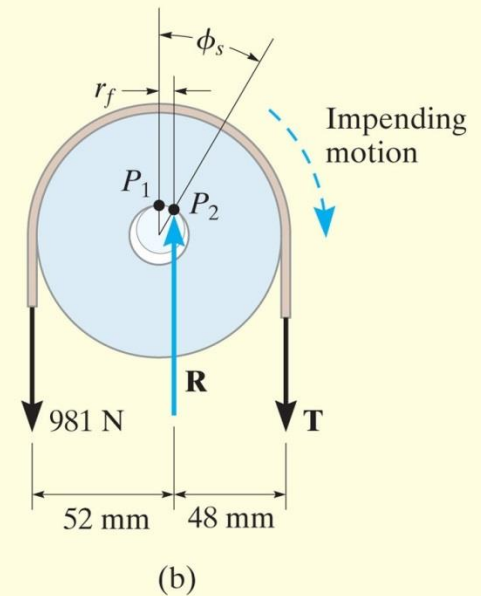
SOLUTION

Part (a). A free-body diagram of the pulley is shown in Fig. 8–24*b*. When the pulley is subjected to belt tensions of 981 N each, it makes contact with the shaft at point P_1 . As the tension T is *increased*, the contact point will move around the shaft to point P_2 before motion impends. From the figure, the friction circle has a radius $r_f = r \sin \phi_s$. Using the simplification that $\sin \phi_s \approx \tan \phi_s \approx \mu_s$ then $r_f \approx r\mu_s = (5 \text{ mm})(0.4) = 2 \text{ mm}$, so that summing moments about P_2 gives

$$\zeta + \Sigma M_{P_2} = 0; \quad 981 \text{ N}(52 \text{ mm}) - T(48 \text{ mm}) = 0$$

$$T = 1063 \text{ N} = 1.06 \text{ kN}$$

Ans.



EXAMPLE 8.10 CONTINUED

If a more exact analysis is used, then $\phi_s = \tan^{-1} 0.4 = 21.8^\circ$. Thus, the radius of the friction circle would be $r_f = r \sin \phi_s = 5 \sin 21.8^\circ = 1.86 \text{ mm}$. Therefore,

$$\zeta + \Sigma M_{P_2} = 0;$$

$$981 \text{ N}(50 \text{ mm} + 1.86 \text{ mm}) - T(50 \text{ mm} - 1.86 \text{ mm}) = 0$$

$$T = 1057 \text{ N} = 1.06 \text{ kN}$$

Ans.

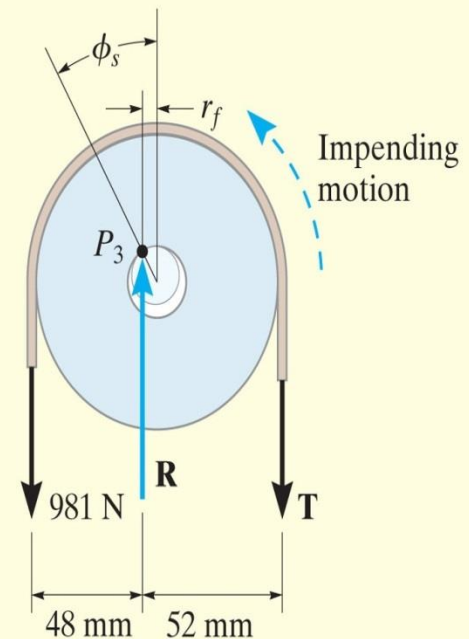
Part (b). When the block is lowered, the resultant force **R** acting on the shaft passes through point as shown in Fig. 8–24c. Summing moments about this point yields

$$\zeta + \Sigma M_{P_3} = 0; \quad 981 \text{ N}(48 \text{ mm}) - T(52 \text{ mm}) = 0$$

$$T = 906 \text{ N}$$

Ans.

NOTE: Using the approximate analysis, the difference between raising and lowering the block is thus 157 N.

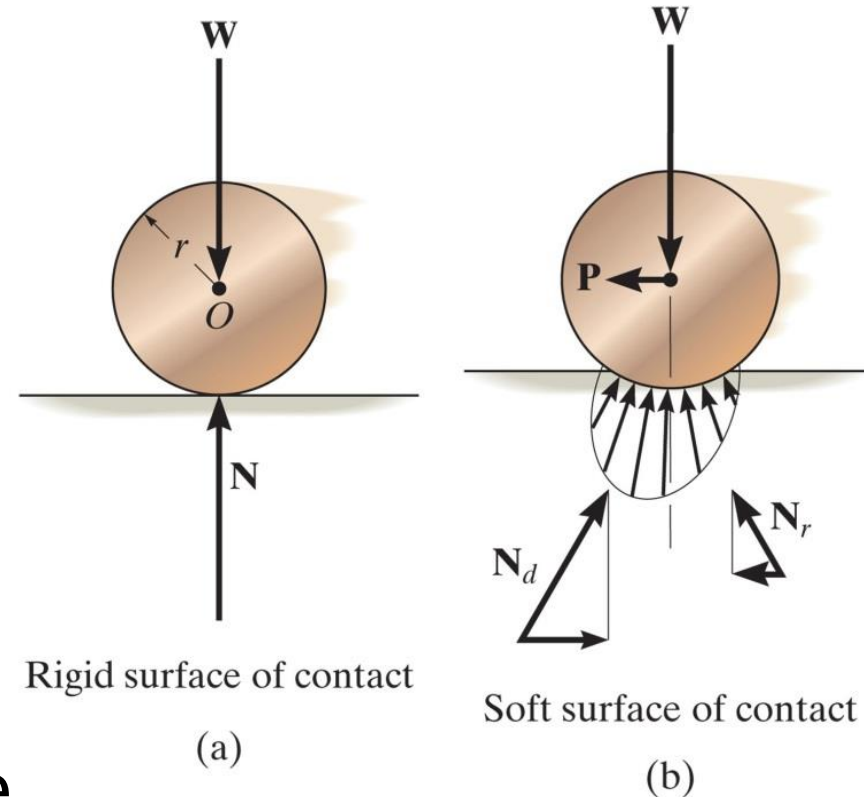


(c)

Fig. 8–24

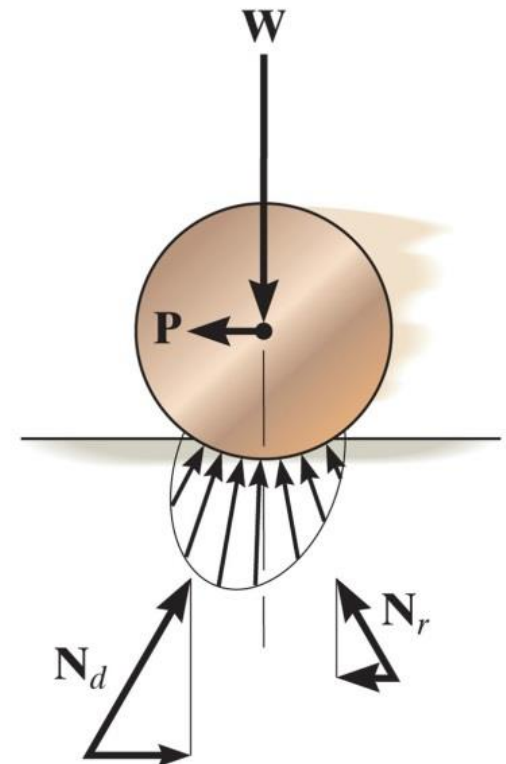
8.8 Rolling Resistance

- ✓ A rigid cylinder rolls at constant velocity along a rigid surface, the normal force is perpendicular to the tangent at the point of contact.
- ✓ However, no material are perfectly rigid.
- ✓ Hard material cylinder will compresses the soft surface underneath it, Fig8-25b



8.8 Rolling Resistance

- ✓ The magnitude of the force of deformation N_d , and its horizontal component is always **greater** than that of restoration N_r .
- ✓ A horizontal driving force P must be applied to the cylinder to maintain the motion.



Soft surface of contact

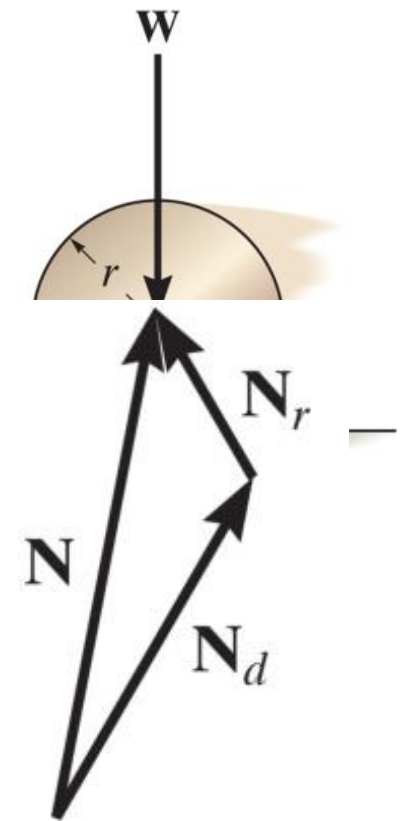
(b)

8.8 Rolling Resistance

- ✓ We consider the resultant of the e pressure, $\mathbf{N} = \mathbf{N}_d + \mathbf{N}_r$, acting on t
- ✓ To keep the cylinder in equilibrium constant rate, $\mathbf{N}$ must be **concurr** driving force $\mathbf{P}$ and the weight $\mathbf{W}$
- ✓ Summing moment about A, give V
Since $\cos \Theta \approx 1 \rightarrow Wa \approx Pr$

or

$$M \approx Rr\mu_k$$

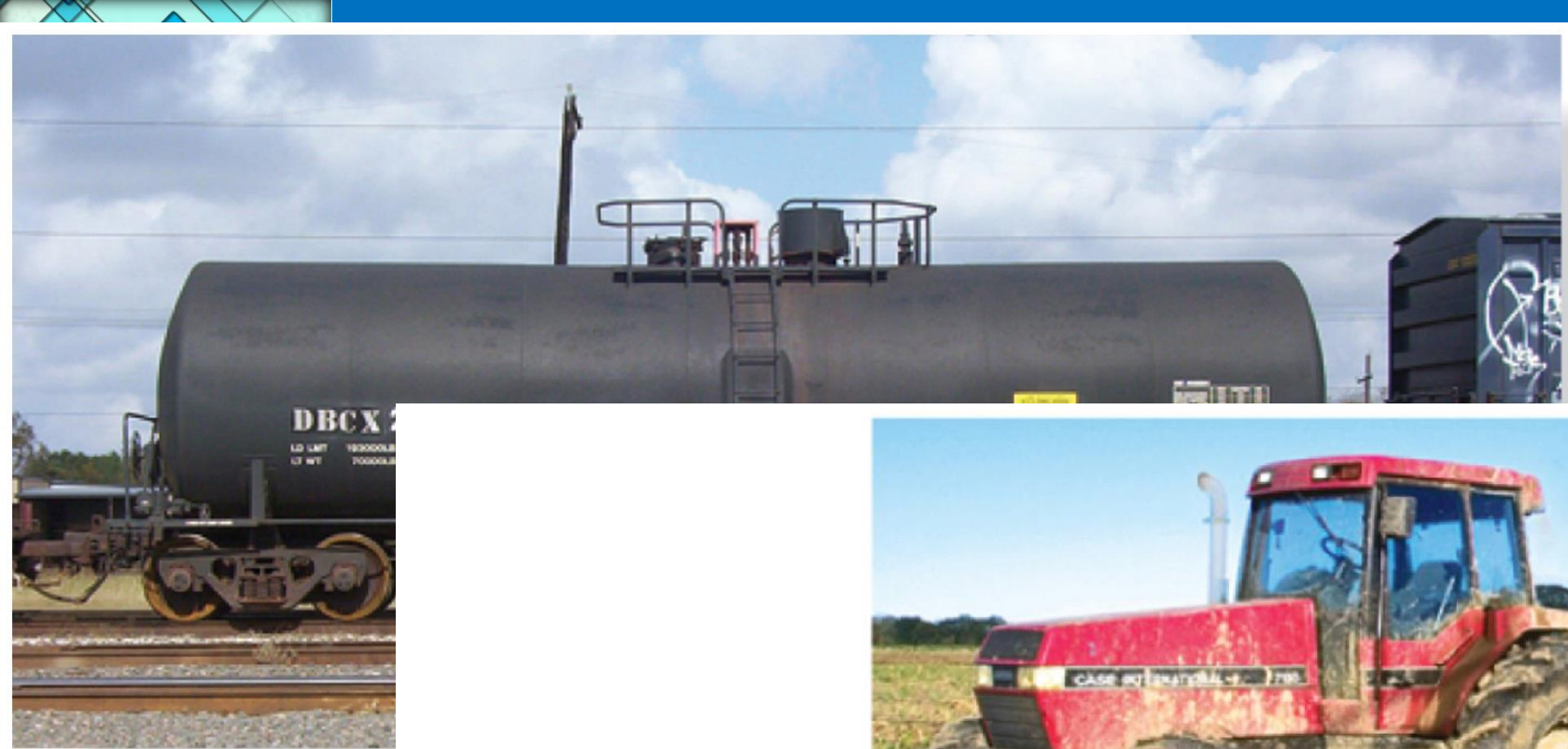


(c)



Figure: 08_PH010

Rolling resistance of railroad wheels on the rails is small since steel is very stiff. By comparison, the rolling resistance of the wheels of a tractor in a wet field is very large.



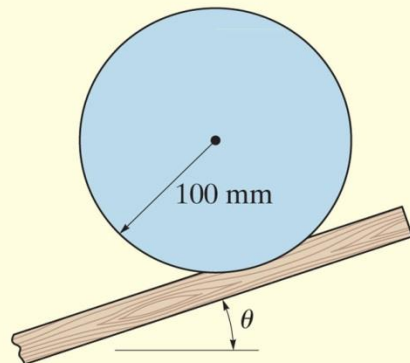
Rolling resistance of
rolling



Figure: 08_PH011
Rolling resistance of railroad wheels on the rails is small since steel is very stiff. By comparison, the rolling resistance of the wheels of a tractor in a wet field is very large.

EXAMPLE 8.11

A 10-kg steel wheel shown in Fig. 8–26*a* has a radius of 100 mm and rests on an inclined plane made of soft wood. If θ is increased so that the wheel begins to roll down the incline with constant velocity when $\theta = 1.2^\circ$, determine the coefficient of rolling resistance.



(a)

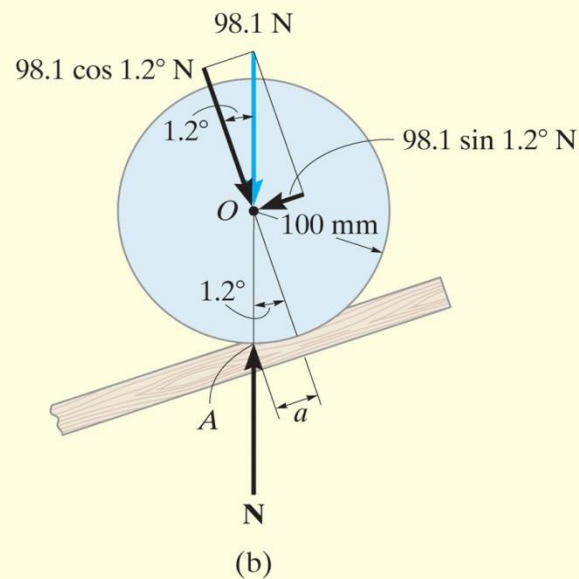


Fig. 8–26

SOLUTION

As shown on the free-body diagram, Fig. 8–26*b*, when the wheel has impending motion, the normal reaction $\mathbf{N}$ acts at point A defined by the dimension a . Resolving the weight into components parallel and perpendicular to the incline, and summing moments about point A , yields

$$\zeta + \Sigma M_A = 0;$$

$$-(98.1 \cos 1.2^\circ \text{ N})(a) + (98.1 \sin 1.2^\circ \text{ N})(100 \cos 1.2^\circ \text{ mm}) = 0$$

Solving, we obtain

$$a = 2.09 \text{ mm}$$

Ans.